

A BILINEAR TWISTED CUBIC MOMENT OF THE RIEMANN ZETA FUNCTION IN THE FIXED-WEIL RANGE $H^{3/2}K \leq T^{1-\eta}$

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ABSTRACT. We evaluate a smoothed bilinear twist of a shifted cubic product of the Riemann zeta function. For two independent divisor-bounded coefficient sequences supported on $h \leq H$ and $k \leq K$, we prove the expected three-term shifted main term, uniformly for shifts of size $O((\log T)^{-1})$, with error

$$O_{\eta,C,\varepsilon}(T^\varepsilon\{KH^{3/2} + T^{1/2}(HK)^{1/2}\})$$

whenever $H^{3/2}K \leq T^{1-\eta}$. The proof fixes one Poisson orientation throughout. Its nonzero frequencies are controlled by finite completion and the Weil bound after exact extraction of (h, km) . The zero frequency is evaluated with a coupled two-variable smoothing. A full-kernel regulator and genuine branchwise Tonelli domains justify the beta integrals; the diagonal and zero-mode residues are assembled before analytic continuation or the final regulator move. A local three-divisor argument proves joint holomorphy when the shifts collide.

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1. INTRODUCTION

Moments of the Riemann zeta function twisted by Dirichlet polynomials are a basic input to mollification and to the study of zeros on the critical line. The twisted second moment is classical [1], while the fourth moment has been evaluated in substantial polynomial-length ranges [6, 2]. The present paper concerns a three-zeta analogue of the twisted second and fourth moments. For smooth w concentrated at height T , consider

$$\sum_{h \leq H} \sum_{k \leq K} \frac{a_h \overline{b_k}}{\sqrt{hk}} \int_{\mathbb{R}} w(t) \zeta\left(\frac{1}{2} + \alpha + it\right) \zeta\left(\frac{1}{2} + \beta + it\right) \zeta\left(\frac{1}{2} + \gamma - it\right) \left(\frac{h}{k}\right)^{it} dt. \quad (1.1)$$

Here a_h and b_k are independent divisor-bounded sequences and the three shifts are $O((\log T)^{-1})$.

Bui introduced the corresponding twisted cubic moment in connection with a three-piece mollifier [4]. The formula in that note was left unproved, and its mollifier application uses two arithmetically different coefficient sequences. We therefore formulate the bilinear problem with independent a_h and b_k . Our main result proves the expected three permutation terms in the fixed-Weil range

$$H^{3/2} K \leq T^{1-\eta}. \quad (1.2)$$

The exact statement, including a uniform error term and the interpretation at colliding shifts, is Theorem 2.1. Its three terms agree with the shifted moment recipe of Conrey, Farmer, Keating, Rubinstein and Snaith [5].

The closest established technology includes the twisted fourth moment of Hughes and Young [6], the quadratic divisor theorem of Bettin, Bui, Li and Radziwiłł [2], and the twisted second-moment work of Bettin, Chandee and Radziwiłł [3]. Those theorems have different zeta-factor or coefficient structures and do not specialize to (1.1). In particular, polarization of a quadratic form with a common structured factor does not create two arbitrary

coefficient sequences in the cubic integral. The proof below instead retains the degenerate equation

$$hn - klm = r$$

coming from the cubic approximate functional equation.

Pliego's asymptotic formulas for special mixed cubic moments [7] involve a fixed one-sided polynomial and likewise do not contain the two arbitrary, asymmetric coefficient sequences in (1.1).

There are two independent analytic issues. For $r \neq 0$, Poisson summation must be carried out only after the common divisor $d = (h, km)$ is extracted; otherwise the modular inverse is not defined. Keeping a single l -Poisson orientation, finite completion and the Weil bound give

$$\mathcal{R}_{\neq 0}^{(l)} \ll_{\varepsilon} T^{\varepsilon} K H^{3/2}.$$

For the zero frequency, formal deletion of the continuous endpoint regions $0 < l < 1$ and $0 < n < 1$ leads to beta integrals outside a common absolute convergence chamber. We resolve this with a coupled (s, u) kernel. Finite variable and Abel regulators are kept while finite contour rectangles are deformed; the rectangle heights tend to infinity before either regulator is removed. Thus the source limit is controlled by the full two-variable kernel, whereas the destination limit is taken in genuine Tonelli domains.

The diagonal and all four zero-mode branches must then be assembled at fixed u . Three pairs of small s -residues cancel, including their finite Euler factors and exact gamma factors. Quantitative holomorphic continuation is applied only to the complete endpoint error after subtracting the full diagonal boundary term. The u -contours are moved afterward. This ordering is what makes the result uniform at $\alpha = \beta = \gamma = 0$.

If $H \leq T^{\theta_1}$ and $K \leq T^{\theta_3}$, the proved open region is

$$\frac{3}{2}\theta_1 + \theta_3 < 1.$$

It includes Bui's boxes $H \leq T^{4/7-\varepsilon_B}$, $K \leq T^{1/4-\varepsilon_B}$ whenever $\varepsilon_B > 3/70$. It does not include the limiting corner $(4/7, 1/4)$; at that corner the pointwise Weil budget is $T^{31/28+o(1)}$. We discuss this exact boundary, without using it in the proof, in Section 10.

Organization. Section 2 states the theorem and records the Euler factors. Section 3 constructs the coupled smoothing and proves its full-contour bounds. Section 4 gives the exact gcd and Poisson decomposition, and Section 5 proves the nonzero-frequency estimate. The endpoint deletion and regulator bridge occupy Section 6. The beta, Euler and archimedean identities are established in Section 7. The fixed- u residue assembly and quantitative continuation are performed in Section 8. The final u -moves and the collision argument appear in Section 9. An audit and reproducibility ledger is given in Appendix A.

2. STATEMENT OF THE THEOREM

Let $C, \eta > 0$ be fixed and let T be sufficiently large in terms of these parameters. Put

$$\Delta = \frac{T}{\log T}.$$

Throughout, $w \in C_c^\infty(\mathbb{R})$ is supported in $[T/4, 2T]$ and satisfies

$$w^{(j)}(t) \ll_j \Delta^{-j} \quad (j \geq 0). \quad (2.1)$$

We make the small-power and seminorm convention explicit. For $r \geq 0$ put

$$\|w\|_{r,\Delta} := \max_{0 \leq j \leq r} \Delta^j \|w^{(j)}\|_\infty, \quad (2.2)$$

$$\|(U, \boldsymbol{\omega})\|_r := \max_{0 \leq j \leq r} \left\{ \|U^{(j)}\|_\infty, \max_{\omega \in \boldsymbol{\omega}} \|\omega^{(j)}\|_\infty \right\}, \quad (2.3)$$

where U is the endpoint cutoff introduced in Section 4 and $\boldsymbol{\omega}$ denotes the five fixed dyadic cutoffs used in Section 5. These auxiliary cutoffs are fixed once and are not additional data in the theorem.

Whenever T^ε or $\ll_\varepsilon$ occurs, $\varepsilon > 0$ denotes the final prescribed small exponent. Auxiliary divisor, localization, and dyadic exponents are chosen sufficiently smaller in terms of this ε , after the finite derivative orders have been fixed. The implied constants and the lower threshold for T may depend on η, C, ε , on the fixed smoothing parameters B_0, M , on fixed contour and integration-by-parts orders, on the coefficient-bound constants at finitely many auxiliary exponents, and on (2.2)–(2.3) for some finite r ; they are uniform in H, K, T and in the shifts in (2.10). The kernels $\mathcal{G}$ and $\mathcal{V}^\pm$ introduce no further independent seminorms, since their bounds are determined by C, B_0, M and the displayed contours. When an $O_D(T^{-D})$ remainder is requested, M and the later integration-by-parts orders may depend on D as specified in Sections 3 and 8; for Theorem 2.1 a sufficiently large D is fixed in terms of the displayed theorem parameters before T varies. In the final theorem B_0, M, U and $\boldsymbol{\omega}$ are proof-internal fixed choices and are suppressed from the $O_{\eta, C, \varepsilon}$ notation; only finitely many scaled seminorms of the external weight w remain. In particular, a harmless T^ε may absorb $(\log T)^{C_M}$ for fixed M , but never a fixed positive power $T^{\delta(M)}$.

For positive integers h, k and complex A, B, C_1 , define first in a region of absolute convergence

$$Z_{A,B,C_1}(h, k) = \sum_{kab=hc} a^{-1/2-A} b^{-1/2-B} c^{-1/2-C_1}. \quad (2.4)$$

The letters a, b, c in this sum are local summation variables and are unrelated to the coefficient sequences in the theorem. To make the continuation explicit, write $g = (h, k)$, $h = gh_0$, $k = gk_0$, and set

$$q_Z = 1 + A + C_1, \quad v_Z = 1 + B + C_1.$$

Then

$$Z_{A,B,C_1}(h, k) = h_0^{1/2+C_1} k_0^{-1/2-C_1} \zeta(q_Z) \zeta(v_Z)$$

$$\times \prod_{p^\nu \parallel h_0} (1 - p^{-qz})(1 - p^{-vz}) \sum_{i+j \geq \nu} p^{-qzi - v z j}. \quad (2.5)$$

We use the right side as the meromorphic continuation of (2.4). Its only shift poles are the two displayed zeta poles.

The exact archimedean factor is

$$X_{A,C_1}(t) = \pi^{A+C_1} \frac{\Gamma((\frac{1}{2} - A - it)/2) \Gamma((\frac{1}{2} - C_1 + it)/2)}{\Gamma((\frac{1}{2} + A + it)/2) \Gamma((\frac{1}{2} + C_1 - it)/2)}. \quad (2.6)$$

For two independent complex sequences (a_h) and (b_k) , put

$$\begin{aligned} \mathcal{I}_{\alpha,\beta,\gamma}(a,b) &= \sum_{h \leq H} \sum_{k \leq K} \frac{a_h \bar{b}_k}{\sqrt{hk}} \int_{\mathbb{R}} w(t) \zeta\left(\frac{1}{2} + \alpha + it\right) \zeta\left(\frac{1}{2} + \beta + it\right) \\ &\quad \times \zeta\left(\frac{1}{2} + \gamma - it\right) \left(\frac{h}{k}\right)^{it} dt. \end{aligned} \quad (2.7)$$

Theorem 2.1 (Fixed-Weil bilinear twisted cubic moment). *Suppose $H, K \geq 1$ and*

$$H^{3/2} K \leq T^{1-\eta}. \quad (2.8)$$

Let

$$|a_h| \ll_{\varepsilon} h^{\varepsilon}, \quad |b_k| \ll_{\varepsilon} k^{\varepsilon} \quad (2.9)$$

for every $\varepsilon > 0$. Uniformly for complex shifts satisfying

$$|\alpha| + |\beta| + |\gamma| \leq \frac{C}{\log T}, \quad (2.10)$$

one has

$$\begin{aligned} \mathcal{I}_{\alpha,\beta,\gamma}(a,b) &= \sum_{h \leq H} \sum_{k \leq K} \frac{a_h \bar{b}_k}{\sqrt{hk}} \int_{\mathbb{R}} w(t) \{ Z_{\alpha,\beta,\gamma}(h,k) \\ &\quad + X_{\alpha,\gamma}(t) Z_{-\gamma,\beta,-\alpha}(h,k) + X_{\beta,\gamma}(t) Z_{\alpha,-\gamma,-\beta}(h,k) \} dt \\ &\quad + O_{\eta,C,\varepsilon} \left(T^{\varepsilon} \{ K H^{3/2} + T^{1/2} (HK)^{1/2} \} \right). \end{aligned} \quad (2.11)$$

The expression in braces is its jointly holomorphic continuation across

$$\alpha + \gamma = 0, \quad \beta + \gamma = 0, \quad \alpha - \beta = 0,$$

including their common collision locus $\alpha = \beta = -\gamma$.

The theorem is formulated with exact gamma ratios. Uniform Stirling asymptotics give

$$X_{A,C_1}(t) = \left(\frac{t}{2\pi} \right)^{-A-C_1} \{1 + O_C(T^{-1})\} \quad (2.12)$$

on the support of w . The individual Z -terms are meromorphic at shift collisions, so this estimate is not applied term by term. Instead, Proposition 8.4 proves that the complete difference between the exact three-term expression and its power-factor analogue is jointly holomorphic and has total contribution $O(T^{\varepsilon}(HK)^{1/2})$. Thus the two exact factors in (2.11) may indeed be

replaced by the corresponding powers used in Bui's formulation, uniformly also at colliding shifts.

Corollary 2.2. *If $H \leq T^{\theta_1}$ and $K \leq T^{\theta_3}$ range over a compact subset of*

$$\frac{3}{2}\theta_1 + \theta_3 < 1,$$

then (2.11) has a power-saving error. More precisely, under (2.8), choosing the harmless T^ε loss with $\varepsilon < \eta/4$ gives an error $O(T^{1-\eta/4})$.

Remark 2.3. No relation between (a_h) and (b_k) is used. Complex conjugation on b_k is only notational; analytically the theorem is bilinear in two independent sequences.

3. COUPLED SMOOTHING AND THE APPROXIMATE FUNCTIONAL EQUATION

Fix $B_0 > 2$ and put $Y = T^{B_0}$. An integer $M \geq 1$ will be chosen after the required final saving has been fixed. Define

$$R_M(u) = \prod_{j=1}^M (1 + u/j) \quad (3.1)$$

and

$$\mathcal{G}(s, u) = e^{s^2} R_M(u) \frac{((1 - \alpha - \gamma - u)^2 - s^2)((1 - \alpha + \beta - u)^2 - s^2)}{(1 - \alpha - \gamma)^2(1 - \alpha + \beta)^2}, \quad (3.2)$$

$$G(s) = \mathcal{G}(s, 0), \quad Q(u) = \mathcal{G}(0, u). \quad (3.3)$$

Thus G is even, $Q(0) = 1$, and the four s -zeros are

$$s = \pm(1 - \alpha - \gamma - u), \quad s = \pm(1 - \alpha + \beta - u). \quad (3.4)$$

The kernel is invariant under $(\beta, \gamma) \mapsto (-\gamma, -\beta)$. In particular, no denominator in (3.2) becomes small in the shift range (2.10).

For shifts B, C_1 , set

$$g_{B, C_1}(s, t) = \pi^{-s} \frac{\Gamma((\frac{1}{2} + B + s + it)/2) \Gamma((\frac{1}{2} + C_1 + s - it)/2)}{\Gamma((\frac{1}{2} + B + it)/2) \Gamma((\frac{1}{2} + C_1 - it)/2)}. \quad (3.5)$$

The factor π^{-s} is included here, so the Mellin argument below is mn , rather than πmn .

Define

$$V_u^+(x, t) = \frac{1}{2\pi i} \int_{(2)} \frac{\mathcal{G}(s, u)}{s} g_{\beta, \gamma}(s, t) x^{-s} ds, \quad (3.6)$$

$$V_u^-(x, t) = \frac{1}{2\pi i} \int_{(2)} \frac{\mathcal{G}(s, u)}{s} g_{-\gamma, -\beta}(s, t) x^{-s} ds. \quad (3.7)$$

Lemma 3.1 (Unnormalized two-piece approximate functional equation). *For every $A_0 > 0$, uniformly for $t \in \text{supp } w$, for the shifts in (2.10), and for $|\Im u| \leq (\log T)^2$, one has*

$$Q(u) \zeta\left(\frac{1}{2} + \beta + it\right) \zeta\left(\frac{1}{2} + \gamma - it\right)$$

$$\begin{aligned}
&= \sum_{m,n \geq 1} \frac{(n/m)^{it}}{m^{1/2+\beta} n^{1/2+\gamma}} V_u^+(mn, t) \\
&\quad + X_{\beta, \gamma}(t) \sum_{m,n \geq 1} \frac{(n/m)^{it}}{m^{1/2-\gamma} n^{1/2-\beta}} V_u^-(mn, t) + O_{A_0}(T^{-A_0}). \tag{3.8}
\end{aligned}$$

More precisely, on any fixed vertical line the error before the u -integral is

$$O_{A_0, M}(T^{-A_0}(1 + |u|)^{2M+4}),$$

and hence remains $O_{A_0}(T^{-A_0})$ after integration against $\Gamma(u)$.

Proof. On $\Re s = 2$, expand the two zeta functions in

$$\frac{1}{2\pi i} \int_{(2)} \frac{\mathcal{G}(s, u)}{s} g_{\beta, \gamma}(s, t) \zeta\left(\frac{1}{2} + \beta + s + it\right) \zeta\left(\frac{1}{2} + \gamma + s - it\right) ds.$$

Move the line to $\Re s = -2$, apply both functional equations, and replace s by $-s$. Evenness of $\mathcal{G}$ produces the dual sum, while the residue at $s = 0$ is $Q(u)$ times the zeta product.

There are also four completed-zeta poles,

$$\frac{1}{2} - \beta - it, \quad -\frac{1}{2} - \beta - it, \quad \frac{1}{2} - \gamma + it, \quad -\frac{1}{2} - \gamma + it.$$

Their imaginary parts have size T , and $e^{s^2} = \exp(-T^2 + O(T))$ there. Their residues and the horizontal integrals are therefore smaller than every power of T . The u -dependence of these errors is polynomial: the only additional factor is the fixed-degree polynomial $\mathcal{G}(s, u)$. Thus, after enlarging A_0 if necessary, the error is $O_{A_0, M}(T^{-A_0}(1 + |u|)^{2M+4})$. On $|\Im u| \leq (\log T)^2$ this gives the stated pointwise uniformity, while on the complementary part of any fixed vertical line Stirling's bound

$$|\Gamma(c + i\tau)| \ll_c (1 + |\tau|)^{c-1/2} e^{-\pi|\tau|/2}$$

makes the polynomial error integrable. This proves both assertions of the lemma. Notice that no division by $Q(u)$ has been made. $\square$

The smoothed long zeta factor is initially defined on an absolutely convergent line by

$$\zeta_\alpha^*\left(\frac{1}{2} + it\right) = \frac{1}{2\pi i} \int_{(1)} \Gamma(u) Q(u) Y^u \zeta\left(\frac{1}{2} + \alpha + u + it\right) du. \tag{3.9}$$

There are two distinct contour operations. To produce a Dirichlet expansion, keep $\Re u = 1$, insert Lemma 3.1, and expand the three Dirichlet series there, with the AFE s -line initially at 2. For finite Dirichlet truncations move only the inverse-Mellin kernel to a fixed line

$$0 < c_u < \frac{1}{4}. \tag{3.10}$$

No pole is crossed. The uniform product bounds in Lemma 3.2 then remove the truncations. Thus the long zeta series is never expanded on the nonconvergent c_u -line.

For comparison with the original zeta function, instead keep the zeta factor in (3.9) unexpanded and move the collective u -line. The identity

$$\Gamma(u)R_M(u) = \frac{\Gamma(u+M+1)}{M!u} \quad (3.11)$$

shows that the poles $-1, \dots, -M$ are killed. Its inverse Mellin weight is

$$W_M(x) = e^{-x} \sum_{j=0}^M \frac{x^j}{j!} = 1 + O_M(x^{M+1}) \quad (x \rightarrow 0). \quad (3.12)$$

The remaining polynomial factors in Q/R_M act by a fixed polynomial in $-x \frac{d}{dx}$ and have value one at $u = 0$. Moving (3.9) to $\Re u = -M - 1/2$ therefore crosses only $u = 0$ and the moving zeta pole $u = 1/2 - \alpha - it$. The latter is exponentially small by Stirling. Hence

$$\zeta_\alpha^*\left(\frac{1}{2} + it\right) = \zeta\left(\frac{1}{2} + \alpha + it\right) + O_{M,C}\left((\log T)^{C_M}(T/Y)^{M+1/2}\right) + O(e^{-cT}). \quad (3.13)$$

For fixed $B_0 > 2$, M can make this error smaller than any prescribed power.

Combining the two Mellin transforms, write

$$\mathcal{V}^\pm(\xi, x, t) = \frac{1}{(2\pi i)^2} \int_{(c_u)} \int_{(2)} \Gamma(u) \frac{\mathcal{G}(s, u)}{s} g_\pm(s, t) \xi^{-u} x^{-s} ds du, \quad (3.14)$$

where $g_+ = g_{\beta, \gamma}$ and $g_- = g_{-\gamma, -\beta}$.

Lemma 3.2 (Full-contour double-weight bounds). *For nonnegative integers i, j, r and every $A_0 > 0$,*

$$\xi^i x^j \left| \partial_\xi^i \partial_x^j \partial_t^r \mathcal{V}^\pm(\xi, x, t) \right| \ll_{i,j,r,A_0,M} (\log T)^{C_M} T^{-r} (1 + \xi)^{-A_0} (1 + x/T)^{-A_0}. \quad (3.15)$$

Every contour used in this bound is a complete infinite vertical line. The same estimate holds, locally uniformly, after replacing (α, β, γ) by external variables (A, B, C_1) with

$$A \in \mathcal{K}_A \Subset \{\Re A < \frac{1}{2} - c_u/2\}, \quad (B, C_1) \in \mathcal{K}_g \Subset \Omega_g,$$

where the two-variable good patch Ω_g is defined in (6.15), and also after the dual substitution $(B, C_1) \mapsto (-C_1, -B)$; the implied constant may depend on these two compact sets.

Proof. Write $s = \sigma + i\tau_s$. The Stirling estimate needed here is made in two regions. If $|\tau_s| \leq |t|/2$, then all four gamma arguments occurring in the two ratios defining $g_\pm$ have imaginary parts comparable with T . Uniform Stirling estimates for the ratios, followed by the corresponding polygamma estimates, give, for every fixed r and every fixed σ used below,

$$|\partial_t^r g_\pm(\sigma + i\tau_s, t)| \ll_{r,\sigma} T^{\sigma-r} (1 + |\tau_s|)^{C_r} \quad (|\tau_s| \leq |t|/2).$$

This is the only region in which a pointwise $T^{\sigma-r}$ estimate is used. On the complementary region, uniform gamma and polygamma bounds away from

their poles give the deliberately coarse estimate

$$|\partial_t^r g_{\pm}(\sigma + i\tau_s, t)| \ll_{r,\sigma} (1 + T + |\tau_s|)^{C_r} \exp\{C_r(T + |\tau_s|)\}.$$

The closed strips swept out below stay to the right of the numerator-gamma poles: for the only left move this follows from $\delta_s < 1/2 - C/\log T$, and the right-moving lines are farther right. Thus the preceding bound is uniform in the real parts occurring on both the vertical lines and their connecting horizontal edges. Since $t \in [T/4, 2T]$, on $|\tau_s| > |t|/2$ the Gaussian satisfies, after enlarging the constants,

$$e^{-\tau_s^2} (1 + T + |\tau_s|)^{C_r} e^{C_r(T + |\tau_s|)} \ll_{r,A_0} T^{-r} e^{-\tau_s^2/2}.$$

For bounded T the same conclusion follows by enlarging the implied constant. Consequently the complete s -contours, including the portions $\tau_s \asymp \pm t$, have an integrable majorant with the required T^{-r} factor. The factor $\Gamma(u)$ gives exponential decay in the other imaginary direction, uniformly after multiplication by the fixed polynomials in s and u . Choose

$$0 < \delta_u < 1, \quad 0 < \delta_s < \frac{1}{2} - C/\log T.$$

For $\xi \geq 1$, move u arbitrarily far right. For $\xi \leq 1$, move it to $\Re u = -\delta_u$; only $u = 0$ is crossed. Similarly, for $x/T \geq 1$ move s right, while for $x/T \leq 1$ move it to $\Re s = -\delta_s$; only $s = 0$ is crossed. In the quadrant where both variables are small, retain both single residues and their common $(u, s) = (0, 0)$ residue, which equals one. In either mixed quadrant, put the residue from the small-variable move on the right-shifted contour of the large variable. This gives

$$\mathcal{V}^{\pm}(\xi, x, t) \ll_{A_0} (1 + \xi)^{-A_0} (1 + x/T)^{-A_0}.$$

Normalized ξ - and x -derivatives insert polynomials in u and s ; these are absorbed by the exponential–Gaussian majorant just obtained. The residues at $u = 0$ and $s = 0$ obey the same bounds (and the common residue is the constant one, whose positive t -derivatives vanish). Thus t -derivatives give the stated T^{-r} on every residue and remaining line. The complete-contour majorants justify every move and avoid endpoint terms from hard height truncation. For the external-variable extension in the statement, A enters only the fixed polynomial kernel $\mathcal{G}(s, u)$ and its two normalizing denominators. On $\mathcal{K}_A \times \mathcal{K}_g$ the real parts of $1 - A - C_1$ and $1 - A + B$ are bounded away from zero, so those denominators have a compact-uniform positive lower bound. The numerator-gamma geometry depends only on B, C_1 and is unchanged. Thus the same contour proof and the same product tails hold with compact-dependent constants. $\square$

4. ARITHMETIC DECOMPOSITION AND EXACT POISSON DENSITY

After inserting Lemma 3.1 into (3.9), the summands in the direct AFE piece, before the outer coefficient weight, are

$$l^{-1/2-\alpha} m^{-1/2-\beta} n^{-1/2-\gamma} \left(\frac{hn}{klm} \right)^{it} \mathcal{V}^+(l/Y, mn, t). \quad (4.1)$$

The dual piece is obtained by

$$(\beta, \gamma, \mathcal{V}^+) \mapsto (-\gamma, -\beta, \mathcal{V}^-) \quad \text{and multiplication by } X_{\beta, \gamma}(t). \quad (4.2)$$

The recovery estimate (3.13) permits us to work with these sums and return to the original cubic moment with an arbitrary power saving.

Put

$$hn - klm = r. \quad (4.3)$$

The terms with $r = 0$ are the direct and dual diagonals. Suppose $r \neq 0$ and set

$$d = (h, km), \quad h = dq, \quad km = dv, \quad r = d\rho. \quad (4.4)$$

Solutions exist only if $d \mid r$. Since $(q, v) = 1$, the equation becomes

$$l \equiv -\rho\bar{v} \pmod{q}. \quad (4.5)$$

Thus every inverse is taken modulo a coprime integer, and Poisson summation in the fixed l -orientation has exact density

$$\frac{1}{q} = \frac{d}{h}. \quad (4.6)$$

We use this same orientation for both zero and nonzero frequencies.

More explicitly, if F is a smooth function of l , then

$$\sum_{l \equiv -\rho\bar{v} \pmod{q}} F(l) = \frac{1}{q} \sum_{a \in \mathbb{Z}} e\left(-\frac{a\rho\bar{v}}{q}\right) \int_{\mathbb{R}} F(y) e\left(-\frac{ay}{q}\right) dy, \quad e(x) = e^{2\pi i x}. \quad (4.7)$$

After the change of variables $x = kmy$, the combined Jacobian and progression density is

$$\frac{1}{qkm} = \frac{d}{hkm}. \quad (4.8)$$

For later use we record the exact zero mode, including the discrete endpoints. Choose $U \in C^\infty(\mathbb{R})$ with $U(x) = 0$ for $x \leq 1/2$ and $U(x) = 1$ for $x \geq 1$. Multiplication by $U(l)U(n)$ does not change the discrete sum. With

$$y = kml, \quad x = hn = y + r, \quad (4.9)$$

the direct zero mode for fixed h, k is

$$\begin{aligned} \mathcal{O}_{0,Y}^+(h, k) &= \int_{\mathbb{R}} w(t) \sum_{m \geq 1} \sum_{\substack{r \neq 0 \\ d \mid r}} d h^{-1/2+\gamma} k^{-1/2+\alpha} m^{-1+\alpha-\beta} \\ &\quad \times \int_{\max(0, -r)}^{\infty} y^{-1/2-\alpha-it} x^{-1/2-\gamma+it} \mathcal{V}^+\left(\frac{y}{kmY}, \frac{mx}{h}, t\right) \end{aligned}$$

$$\times U\left(\frac{y}{km}\right) U\left(\frac{x}{h}\right) dy dt. \quad (4.10)$$

The outside factor $a_h \overline{b_k} / \sqrt{hk}$ is not included in (4.10). Formula (4.10) follows directly from (4.6) after changing dl to $dy/(km)$; in particular, the factor d has not been suppressed. The dual zero mode is obtained by (4.2).

Let $\tilde{\mathcal{O}}_{0,Y}^+$ be (4.10) with both U factors deleted, initially wherever the resulting integrals and sums converge absolutely and subsequently by the continuation constructed in Section 6. Define $\tilde{\mathcal{O}}_{0,Y}^-$ in the same way.

For the nonzero-frequency analysis, decompose all five arithmetic variables into dyadic blocks

$$h \asymp H_0, \quad k \asymp K_0, \quad l \asymp L_0, \quad m \asymp M_0, \quad n \asymp N_0$$

and put

$$S = H_0 N_0 + K_0 L_0 M_0. \quad (4.11)$$

Repeated integration by parts in t shows that a non-negligible block must satisfy

$$H_0 N_0 \asymp K_0 L_0 M_0, \quad |r| \ll ST^{-1+\varepsilon}. \quad (4.12)$$

Consequently

$$S \asymp H_0 N_0 \asymp K_0 L_0 M_0 \asymp (H_0 K_0 L_0 M_0 N_0)^{1/2}. \quad (4.13)$$

The AFE conductor also gives

$$M_0 N_0 \ll T^{1+\varepsilon}, \quad (4.14)$$

while the long smoothing has $L_0 \ll YT^\varepsilon$. The logarithmic number of blocks is absorbed into T^ε .

5. THE NONZERO POISSON FREQUENCIES

We prove the estimate that determines the range of Theorem 2.1. The argument uses only finite completion and the classical Weil bound [8]; it does not correlate the two coefficient sequences.

Throughout this section the convention following (2.1) is in force. Thus the constants behind T^ε may depend on the fixed C, B_0, M , on finitely many seminorms (2.2)–(2.3), and on the fixed completion order, but not on the dyadic block parameters or on H, K, T . The parameter η enters only when (2.8) is invoked in the global summation. The auxiliary ε_0 used below is chosen only after the completion order is known and is then absorbed into the final ε ; it is not an additional uniformity parameter.

For a dyadic block as in Section 4, the t -integral and the change of variables in (4.8) must be kept together. Along $hn - klm = r$, the phase is $x/(x+r)$, and

$$x \frac{d}{dx} \left(t \log \frac{x}{x+r} \right) \ll \frac{T|r|}{S} \ll T^\varepsilon. \quad (5.1)$$

Consequently the Poisson integral has size TS , with normalized derivatives bounded by $TS T^\varepsilon$. The factor T here is essential.

For $d = (h, km)$ as in (4.4), integration by parts in the Poisson integral restricts a nonzero Fourier frequency to

$$0 < |a| \leq A_d, \quad A_d \ll \frac{H_0 K_0 M_0}{dS} T^\varepsilon \asymp \frac{S}{dL_0 N_0} T^\varepsilon. \quad (5.2)$$

Also

$$0 < |\rho| \ll \frac{R_0}{d}, \quad R_0 = ST^{-1+\varepsilon}. \quad (5.3)$$

If either upper bound is less than one, the corresponding block is empty.

Lemma 5.1 (Incomplete reciprocal sum). *Let $q, c \geq 1$, and let F be supported in an interval of length X with $F^{(j)} \ll_j ZX^{-j}$. Then*

$$\sum_{(u,q)=1} F(u) e\left(\frac{c\bar{u}}{q}\right) \ll_\varepsilon Zq^\varepsilon \left\{ Xq^{-1/2}(c,q)^{1/2} + q^{1/2}(c,q)^{1/2} \right\}. \quad (5.4)$$

Moreover,

$$\sum_{q \gtrsim Q} (c,q)^{1/2} \ll_\varepsilon Qc^\varepsilon. \quad (5.5)$$

Proof. With the Fourier-transform convention compatible with $e(x) = e^{2\pi i x}$, finite completion gives

$$\sum_{(u,q)=1} F(u) e\left(\frac{c\bar{u}}{q}\right) = \frac{1}{q} \sum_{n \in \mathbb{Z}} S(c, n; q) \hat{F}(n/q),$$

where

$$S(c, n; q) = \sum_{u \bmod q}^* e\left(\frac{c\bar{u} + nu}{q}\right).$$

For every fixed $A \geq 2$, repeated integration by parts gives

$$|\hat{F}(\xi)| \ll_A ZX(1 + X|\xi|)^{-A}, \quad \sum_{n \in \mathbb{Z}} |\hat{F}(n/q)| \ll_A Z(X + q).$$

The Weil bound $|S(c, n; q)| \leq \tau(q)(c, n, q)^{1/2} q^{1/2}$, together with $(c, n, q)^{1/2} \leq (c, q)^{1/2}$ for every n , therefore proves (5.4), uniformly for both $X \leq q$ and $X > q$. Thus the factor $(c, q)^{1/2}$ occurs in both the $Xq^{-1/2}$ term and the $q^{1/2}$ term. The term $n = 0$ is the Ramanujan sum and obeys the same majorant. Finally, $(c, q)^{1/2} \leq \sum_{e|(c,q)} e^{1/2}$ gives

$$\sum_{q \gtrsim Q} (c, q)^{1/2} \leq \sum_{e|c} e^{1/2} \#\{q \asymp Q : e \mid q\} \ll Q \sum_{e|c} e^{-1/2} \ll_\varepsilon Qc^\varepsilon,$$

which is (5.5). $\square$

For the coefficient block put

$$\mathcal{A}_{H_0} = \max_{h \gtrsim H_0} |a_h|, \quad \mathcal{B}_{K_0} = \sum_{k \gtrsim K_0} |b_k|. \quad (5.6)$$

Proposition 5.2 (A fixed-orientation block estimate). *The contribution of the nonzero frequencies obtained by Poisson summation in l satisfies*

$$E^{(l)}(H_0, K_0, L_0, M_0, N_0) \ll_\varepsilon T^\varepsilon \mathcal{A}_{H_0} \mathcal{B}_{K_0} H_0^{1/2} (H_0 + M_0). \quad (5.7)$$

Proof. Choose the dyadic partition using fixed functions

$$\omega_H, \omega_K, \omega_L, \omega_M, \omega_N \in C_c^\infty((1/2, 3)).$$

All their derivatives are bounded independently of the block parameters. We first expose the smooth weight to which Lemma 5.1 is applied.

Fix d, a, ρ and k , and write

$$g = (d, k), \quad d = g\delta, \quad k = gk_1, \quad m = \delta m_1. \quad (5.8)$$

The equality $(d, k) = g$ implies $(\delta, k_1) = 1$: for every prime p , the minimum of the p -adic valuations of d/g and k/g is zero. Since $d \mid km$, it follows that $\delta \mid m$, so m_1 is an integer. If $h = dq$, then

$$(h, km) = d \iff (q, k_1 m_1) = 1, \quad \frac{km}{d} = k_1 m_1. \quad (5.9)$$

All substitutions are uniquely determined by d, h, k, m ; conversely the right side implies the original gcd condition. Thus this is a bijective parametrization, with neither omission nor repeated counting.

Put $r = d\rho$ and, after the change of variables $x = kml$ from (4.8), set

$$\ell_x = \frac{x}{km}, \quad n_x = \frac{x + d\rho}{h}. \quad (5.10)$$

For the direct AFE piece define the exact normalized amplitude

$$\begin{aligned} \mathcal{W}_{h,k,m,d\rho}^+(x, t) &= \frac{(H_0 K_0 L_0 M_0 N_0)^{1/2} H_0 K_0 M_0}{hkm (hk\ell_x m n_x)^{1/2}} \\ &\times \omega_H(h/H_0) \omega_K(k/K_0) \omega_L(\ell_x/L_0) \\ &\times \omega_M(m/M_0) \omega_N(n_x/N_0) \ell_x^{-\alpha} m^{-\beta} n_x^{-\gamma} w(t) \left(\frac{x + d\rho}{x} \right)^{it} \\ &\times \mathcal{V}^+\left(\frac{x}{kmY}, \frac{m(x + d\rho)}{h}, t \right). \end{aligned} \quad (5.11)$$

The cutoffs make the integrand zero unless $x > 0$ and $x + d\rho > 0$. The dual amplitude is obtained by the exact substitution (4.2); its additional factor $X_{\beta,\gamma}(t)$ is independent of m_1 and is $T^{O(1/\log T)}$ on the support of w .

For either AFE sign let

$$\mathfrak{F}_{d,a,\rho,q,k}^\pm(m_1) = \frac{1}{TS} \int_{\mathbb{R}} \int_{\mathbb{R}} \mathcal{W}_{dq,k,\delta m_1,d\rho}^\pm(x, t) e\left(-\frac{ax}{qk\delta m_1}\right) dx dt. \quad (5.12)$$

Combining (4.1), (4.7), and (4.8), the contribution for fixed d, a, ρ, q, k, m_1 , before the reciprocal phase is summed, is exactly

$$\frac{dT S}{(H_0 K_0 L_0 M_0 N_0)^{1/2} H_0 K_0 M_0} a_{dq} \bar{b}_k \mathfrak{F}_{d,a,\rho,q,k}^\pm(m_1) e\left(-\frac{a\rho \bar{k}_1 m_1}{q}\right), \quad (5.13)$$

up to the common sign convention for the two exponentials. The factor d is precisely the numerator of the density $d/(hkm)$ and has not been hidden in the smooth weight.

We now verify the completion hypotheses. Fix an auxiliary $0 < \varepsilon_0 < \varepsilon$ and invoke the localization, frequency truncation, and divisor estimates in this proof with ε_0 in place of ε ; its size will be fixed after the finite completion order is known. The support is contained in a fixed multiple of an interval of length

$$X_m = \frac{M_0}{\delta} = \frac{M_0 g}{d}. \quad (5.14)$$

If this interval contains an integer, then $X_m \gg 1$; otherwise the sum is empty. On the x -support one has $x \asymp K_0 L_0 M_0 \asymp S$, where $S = H_0 N_0 + K_0 L_0 M_0$ is fixed throughout the block. Put $D_x = x \partial_x$. Since $|d\rho|/x \ll T^{-1+\varepsilon_0}$, the identity

$$\log \frac{x}{x+d\rho} = - \int_0^{d\rho} \frac{dv}{x+v}$$

gives, for every fixed $j \geq 0$,

$$\left| D_x^j \left[t \log \frac{x}{x+d\rho} \right] \right| \ll_j \frac{T|d\rho|}{S} \ll T^{\varepsilon_0}. \quad (5.15)$$

The segment $x+v$ remains comparable with S , also when $d\rho < 0$. Repeated differentiation of the exponential therefore costs only $T^{O_j(\varepsilon_0)}$.

At fixed x the t -oscillation is independent of m_1 , while

$$m_1 \partial_{m_1} \ell_x = -\ell_x, \quad m_1 \partial_{m_1} n_x = 0, \quad m_1 \partial_{m_1} \left[t \log \frac{x}{x+d\rho} \right] = 0. \quad (5.16)$$

If the moving support is instead fixed by $x = k\delta m_1 L_0 y$, then

$$m_1 \partial_{m_1} |_y = m_1 \partial_{m_1} |_x + D_x,$$

so (5.15) gives the same conclusion. In particular, this change does not differentiate the fixed block-scale S . For

$$\xi = \frac{x}{kmY}, \quad \chi = \frac{m(x+d\rho)}{h},$$

one has

$$m_1 \partial_{m_1} \xi = -\xi, \quad m_1 \partial_{m_1} \chi = \chi. \quad (5.17)$$

Thus (3.15) controls every derivative falling on $\mathcal{V}^\pm$. Derivatives of the dyadic cutoffs cost X_m^{-1} , and all algebraic powers have bounded logarithmic derivatives on their supports. Finally,

$$\left| \frac{ax}{qkm} \right| \ll \frac{|a|dS}{H_0 K_0 M_0} \ll T^{\varepsilon_0} \quad (5.18)$$

by (5.2); hence every fixed number of derivatives of the inner Poisson Fourier phase is also $T^{O_j(\varepsilon_0)}$.

The x -support has length $O(S)$ and the t -support length $O(T)$. Differentiating under the integrals in (5.12) therefore yields, for every fixed $j \geq 0$,

$$\left| (m_1 \partial_{m_1})^j \mathfrak{F}_{d,a,\rho,q,k}^\pm(m_1) \right| \ll_j T^{O_j(\varepsilon_0)}, \quad \left| \frac{d^j}{dm_1^j} \mathfrak{F}_{d,a,\rho,q,k}^\pm(m_1) \right| \ll_j T^\varepsilon X_m^{-j}. \quad (5.19)$$

Here the localization loss ε_0 is chosen sufficiently small in terms of the final ε and the fixed number of derivatives used in completion. No derivative order depends on T .

Since $(q, k_1) = 1$, the coefficient in the reciprocal phase may be taken as $c = \pm a \rho \overline{k_1} \pmod{q}$, for which $(c, q) = (a\rho, q)$. Applying Lemma 5.1 to the actual weight gives

$$\begin{aligned} & \sum_{\substack{m_1 \asymp M_0 g/d \\ (m_1, q)=1}} \mathfrak{F}_{d,a,\rho,q,k}^\pm(m_1) e\left(\frac{\pm a \rho \overline{k_1} m_1}{q}\right) \\ & \ll T^\varepsilon \left\{ \frac{M_0 g}{d} q^{-1/2} (a\rho, q)^{1/2} + q^{1/2} (a\rho, q)^{1/2} \right\}. \end{aligned} \quad (5.20)$$

Put $Q_d = H_0/d$. By (5.5), restriction to $(q, k_1) = 1$ can only decrease the majorant. Applying (5.5) separately to the two terms in (5.20) gives

$$\frac{M_0 g}{d} Q_d^{-1/2} \sum_{q \asymp Q_d} (a\rho, q)^{1/2} \ll_\varepsilon \frac{M_0 g}{d} Q_d^{1/2} |a\rho|^\varepsilon$$

and

$$Q_d^{1/2} \sum_{q \asymp Q_d} (a\rho, q)^{1/2} \ll_\varepsilon Q_d^{3/2} |a\rho|^\varepsilon.$$

Consequently,

$$\begin{aligned} & \sum_{q \asymp Q_d} \left(\frac{M_0 g}{d} q^{-1/2} + q^{1/2} \right) (a\rho, q)^{1/2} \\ & \ll T^\varepsilon \left(\frac{M_0 g H_0^{1/2}}{d^{3/2}} + \frac{H_0^{3/2}}{d^{3/2}} \right). \end{aligned} \quad (5.21)$$

Indeed these are respectively $X_m Q_d^{1/2}$ and $Q_d^{3/2}$. The factor $|a\rho|^\varepsilon$ is absorbable because all retained dyadic parameters are bounded by a fixed power of T ; the auxiliary exponent is chosen before relabelling the total loss as T^ε .

Take the supremum norm of a_{dq} and sum over k . The sets $\{k : (d, k) = g\}$ partition the k -block. The second term of (5.21) is independent of g , and in the first we use only $g \leq d$. Therefore

$$\sum_{\substack{h \asymp H_0, k \asymp K_0, m \asymp M_0 \\ (h, km)=d}} a_h \overline{b_k} \mathfrak{F}_{d,a,\rho,h/d,k}^\pm(m/\delta) e\left(\frac{\pm a \rho \overline{km/d}}{h/d}\right)$$

$$\ll T^\varepsilon \mathcal{A}_{H_0} \mathcal{B}_{K_0} \left(\frac{H_0^{3/2}}{d^{3/2}} + \frac{M_0 H_0^{1/2}}{d^{1/2}} \right). \quad (5.22)$$

We finish by multiplying every scalar factor. After the density numerator d is removed from (5.13), the original Dirichlet normalization, the remaining progression density and Jacobian, and the actual t, x -integral size give

$$\mathcal{C}_0 = \frac{TS}{(H_0 K_0 L_0 M_0 N_0)^{1/2} H_0 K_0 M_0}. \quad (5.23)$$

More explicitly, the original normalization is $(H_0 K_0 L_0 M_0 N_0)^{-1/2}$, the progression density is d/H_0 , the $x = kml$ Jacobian costs $(K_0 M_0)^{-1}$, and the Poisson integral has size TS . For fixed d , the density numerator, the number of frequencies, and the number of nonzero shifts contribute

$$d \cdot \frac{A_0}{d} \cdot \frac{R_0}{d} = \frac{A_0 R_0}{d}, \quad A_0 = \frac{H_0 K_0 M_0}{S} T^{\varepsilon_0}, \quad R_0 = \frac{S}{T} T^{\varepsilon_0}. \quad (5.24)$$

Multiplying (5.22)–(5.24) and extending the d -sum to all positive integers gives

$$\begin{aligned} E^{(l)}(H_0, K_0, L_0, M_0, N_0) &\ll T^\varepsilon \mathcal{C}_0 A_0 R_0 \mathcal{A}_{H_0} \mathcal{B}_{K_0} \\ &\times \left(H_0^{3/2} \sum_{d \geq 1} d^{-5/2} + M_0 H_0^{1/2} \sum_{d \geq 1} d^{-3/2} \right). \end{aligned} \quad (5.25)$$

Both d -series converge. Finally, by (4.13),

$$\mathcal{C}_0 A_0 R_0 = \frac{S}{(H_0 K_0 L_0 M_0 N_0)^{1/2}} T^{O(\varepsilon_0)} \asymp T^{O(\varepsilon_0)}. \quad (5.26)$$

Choosing ε_0 in terms of the final ε proves (5.7). $\square$

Proposition 5.3 (Nonzero-frequency estimate). *With the fixed l -Poisson orientation,*

$$\mathcal{R}_{\neq 0}^{(l)} \ll_\varepsilon T^\varepsilon \|a\|_\infty H^{3/2} \sum_{k \leq K} |b_k|. \quad (5.27)$$

In particular, under (2.9),

$$\mathcal{R}_{\neq 0}^{(l)} \ll_\varepsilon T^\varepsilon K H^{3/2}. \quad (5.28)$$

Proof. For a nonempty nonzero-frequency block, $r = d\rho \neq 0$ and (4.12) imply

$$S \gg dT^{1-\varepsilon}. \quad (5.29)$$

Since $N_0 \asymp S/H_0$ and $M_0 N_0 \ll T^{1+\varepsilon}$,

$$M_0 \ll \frac{H_0 T^{1+\varepsilon}}{S} \ll \frac{H_0}{d} T^{2\varepsilon} \leq H_0 T^{2\varepsilon}. \quad (5.30)$$

Substitution in (5.7), followed by the logarithmic dyadic sums and the two AFE pieces, gives (5.27). The divisor bounds then give (5.28). $\square$

Remark 5.4. Poisson summation in another variable gives a separately valid estimate for a different decomposition, but its zero frequency is different. We do not take a minimum over Poisson orientations. The fixed orientation used in Proposition 5.3 is exactly the one whose zero mode is evaluated in the following sections.

6. ENDPOINT DELETION AND THE FULL-KERNEL BRIDGE

The zero mode (4.10) is initially a continuous integral with the discrete restrictions $l, n \geq 1$ retained by U . We first delete the artificial regions $0 < l < 1$ and $0 < n < 1$ on a genuine convergence patch. We then construct the exact continuation needed to reach all small shifts.

6.1. Fourier decay and the endpoint lemma.

Lemma 6.1 (Fourier-conductor bound). *Uniformly for $x, y > 0$, $x \neq y$, and every $A_0, J > 0$,*

$$\begin{aligned} & \left| \int_{\mathbb{R}} w(t)(x/y)^{it} \mathcal{V}^{\pm} \left(\frac{y}{kmY}, \frac{mx}{h}, t \right) dt \right| \\ & \ll_{A_0, J} T(\log T)^{C_M} \left(1 + \frac{y}{kmY} \right)^{-A_0} \\ & \quad \times \left(1 + \frac{mx}{hT} \right)^{-A_0} (1 + \Delta |\log(x/y)|)^{-J}. \end{aligned} \quad (6.1)$$

Proof. Integrate by parts J times in t and use (2.1) and (3.15). Every derivative costs at most Δ^{-1} , and the support of w has length $O(T)$. $\square$

Proposition 6.2 (Endpoint deletion on the good half-space). *Assume (2.8). For every $D > 0$, after summing over $h \leq H$, $k \leq K$ with the outside coefficient weights, one has*

$$\mathcal{O}_{0,Y}^{\pm} - \tilde{\mathcal{O}}_{0,Y}^{\pm} = O_D(T^{-D}) \quad (6.2)$$

when

$$\Re(\beta - \gamma) \geq 0, \quad |\Re \alpha| + |\Re \beta| + |\Re \gamma| \leq C/\log T. \quad (6.3)$$

The source expressions retain the complete contours in (3.14).

Proof. It suffices to treat

$$\mathcal{E}_l = \{y < km\}, \quad \mathcal{E}_n = \{x < h\}. \quad (6.4)$$

Put $\delta_0 = \eta/4$ and $\varepsilon_0 = \eta/8$. On $\mathcal{E}_l$, suppose simultaneously that

$$|\log(x/y)| \leq T^{-1+\delta_0}, \quad \frac{mx}{hT} \leq T^{\varepsilon_0}. \quad (6.5)$$

Then $x \asymp y$, and, with $r = d\rho$, $h = dq$, $km = dv$,

$$y \gg d|\rho|T^{1-\delta_0}, \quad v \gg |\rho|T^{1-\delta_0}, \quad m|\rho| \ll qT^{\delta_0+\varepsilon_0}. \quad (6.6)$$

Since $v \leq km$ and $q \leq h \leq H$, this forces

$$HK \gg T^{1-2\delta_0-\varepsilon_0} = T^{1-5\eta/8}. \quad (6.7)$$

But (2.8) and $H \geq 1$ give $HK \leq T^{1-\eta}$, a contradiction for large T . Every point of $\mathcal{E}_l$ therefore has either Fourier saving $|\log(x/y)| > T^{-1+\delta_0}$ or conductor saving $mx/(hT) > T^{\varepsilon_0}$. The orders A_0, J in (6.1) absorb every remaining polynomial factor.

On $\mathcal{E}_n$, $0 < x < h$ and $r = d\rho \neq 0$ imply

$$|\log(x/y)| \gg \min(1, |r|/h) \gg 1/H. \quad (6.8)$$

The main range gives $H \leq T^{2(1-\eta)/3}$, and hence

$$\Delta |\log(x/y)| \gg T^{(1+2\eta)/3} / \log T. \quad (6.9)$$

Repeated integration by parts again gives an arbitrary power saving, once the endpoint mass is shown to be only polynomial.

We record the singular-layer summation because it is where the gcd density is needed. For fixed m and a dyadic l -scale, the values

$$n = \frac{kml + d\rho}{h}$$

form a mesh of spacing d/h , exactly matching (4.6). A boundary layer meeting $n = 0$ has $kml \asymp d|\rho|$. On writing $n = (d/h)x$ and $x = |\rho|z$, the square-root Jacobian is

$$|\rho|^{-1/2} \int_0^{O(|\rho|)} x^{-1/2}(\dots) dx = \int_0^{O(1)} z^{-1/2}(\dots) dz. \quad (6.10)$$

Thus no unrecorded sum over ρ and no lost $|\rho|^{-1/2}$ occurs.

For the potentially worst case $r < 0$, $x < h$, partial summation in $r = d\rho$ and then in m gives, up to T^ε ,

$$\sum_{\rho \geq 1} y^{-1/2-\Re\alpha} \left(1 + \frac{y}{kmY}\right)^{-A_0} \ll d^{-1}(kmY)^{1/2-\Re\alpha}, \quad (6.11)$$

$$\sum_{m \geq 1} m^{-1/2-\Re\beta} \left(1 + \frac{mx}{hT}\right)^{-A_0} \ll (hT/x)^{1/2-\Re\beta}. \quad (6.12)$$

After multiplication by $x^{-1/2-\Re\gamma}$, the lower-end behavior is

$$x^{-1+\Re(\beta-\gamma)}. \quad (6.13)$$

It is integrable in (6.3); on the boundary, the last factor in (6.1) makes the logarithmic endpoint integrable for $J > 1$. All remaining sums have fixed polynomial size and are absorbed by (6.9). This proves (6.2). $\square$

The ordinary dominated-convergence proof stops at $\Re(\beta - \gamma) = 0$: its far dyadic majorant has residual size

$$M_1^{\Re(\gamma-\beta)} \{1 + \log(M_1/T)\}^{-J} \quad (M_1 \gg T). \quad (6.14)$$

We therefore continue only the complete assembled error, rather than a single endpoint-deleted branch.

6.2. Source normality with the full kernel. In this subsection use external shifts A, B, C_1 in place of α, β, γ in the kernel. The continuation requires only one nonempty patch. For definiteness take $\kappa_0 = 1$, $c_1 = 5$, and define the following domain in the two variables (B, C_1) :

$$\Omega_g = \left\{ \Re B > \frac{\kappa_0}{\log T}, \quad \Re C_1 < -\frac{\kappa_0}{\log T}, \quad |B| + |C_1| < \frac{c_1}{\log T} \right\} \subset \mathbb{C}^2. \quad (6.15)$$

The initial three-variable good patch additionally has $|A| + |B| + |C_1| < c_1/\log T$. After fixing $(B, C_1) \in \Omega_g$, however, we let A vary independently in the connected strip

$$\Re A < \frac{1}{2} - \frac{c_u}{2}. \quad (6.16)$$

Lemma 6.3 (Full-kernel source normality). *For each of the four pairs*

(direct or dual AFE sign, $\operatorname{sgn} \rho$),

the endpoint-deleted expressions, with the finite regulators introduced below, converge locally uniformly in the strip (6.16), uniformly for (B, C_1) on compact subsets of (6.15). Their limits are holomorphic in A , as are any fixed number of A -derivatives.

Proof. Keep the complete kernel $\mathcal{V}(l/Y, mn, t)$ and do not freeze its s variable. For the exact-cutoff source, $l, n \geq 1/2$. If $F(n)$ is the nonnegative product-tail majorant from Lemma 3.2, then on either half-mesh of spacing $\delta = d/h$,

$$\delta \sum_{\rho: n \geq 1/2} F(n) \ll \int_{1/2}^{\infty} F(v) dv + \delta \sup_{v \geq 1/2} F(v). \quad (6.17)$$

This gives, before the finite outer sums, a fixed logarithmic power times

$$T \int_{1/2}^{\infty} l^{-1/2-a_0} (1 + l/Y)^{-P} dl \sum_{m \geq 1} m^{-1/2-b_0} \times \left\{ \int_{1/2}^{\infty} n^{-1/2-c_0} (1 + mn/T)^{-Q} dn + (1 + m/T)^{-Q} \right\}. \quad (6.18)$$

where $a_0 = \Re A$, $b_0 = \Re B$, and $c_0 = \Re C_1$. On compact subsets of (6.16), this is polynomial in T and locally uniform; the m -sum containing the n -integral is $O_{\mathcal{K}_g}(\log T)$ on a compact $\mathcal{K}_g \Subset \Omega_g$, whereas the mesh-supremum term is

$$\sum_{m \geq 1} m^{-1/2-b_0} (1 + m/T)^{-Q} \ll_{Q, \mathcal{K}_g} T^{1/2-b_0}. \quad (6.19)$$

Thus the complete expression is still of fixed polynomial size, which is all that is used here. The same statements hold after the dual substitution $(B, C_1) \mapsto (-C_1, -B)$.

For completeness, consider the worst $n = 0$ discrepancy with $r = -d\rho$, $z = x/(d\rho) \leq 1$, and restore the outside coefficient weight. Its full algebraic prefactor is

$$|a_h b_k| d^{1-a_0-c_0} h^{-1+c_0} k^{-1+a_0} m^{-1+a_0-b_0} \rho^{-a_0-c_0} \\ \times \int_0^1 z^{-1/2-c_0} \left(1 + \frac{md\rho z}{hT}\right)^{-Q} \left(1 + \frac{d\rho}{kmY}\right)^{-P} (1 + |\log z|)^{-J} dz. \quad (6.20)$$

For $X = md\rho/(hT)$,

$$\int_0^1 z^{-1/2-c_0} (1+Xz)^{-Q} (1+|\log z|)^{-J} dz \ll (1+X)^{-1/2+c_0} \{1+\log(2+X)\}^{-J}. \quad (6.21)$$

In the range $X > 1$, summing m and ρ gives at most

$$\left(\frac{hT}{d}\right)^{1/2-c_0} \left(\frac{kY}{d}\right)^{1/2-a_0} \sum_{m \geq 1} m^{-1-b_0+c_0}. \quad (6.22)$$

The last sum is $O_{\kappa_0}(\log T)$. Multiplication by the prefactor in (6.20) cancels every power of d and leaves

$$|a_h b_k| (hk)^{-1/2} T^{1/2-c_0} Y^{1/2-a_0} \sum_{m \geq 1} m^{-1-b_0+c_0}. \quad (6.23)$$

After the h, k sums and the Fourier prefactor, the $X > 1$ discrepancy is

$$\ll T \Delta^{-J} T^{1/2-c_0} Y^{1/2-a_0} (HK)^{1/2} T^\epsilon \log T, \quad (6.24)$$

and is $O_N(T^{-N})$ after increasing J . This exact cancellation is uniform over all gcd blocks. The range $X \leq 1$ has $m\rho \leq hT/d$ and therefore only polynomial mass on compact external A -strips. For $z > 1$, $x < h = dq$ gives

$$\left| \log \frac{x}{x+d\rho} \right| = \log(1+z^{-1}) \gg q^{-1} \geq H^{-1}. \quad (6.25)$$

In both cases an arbitrarily high power in the Fourier bound absorbs the polynomial mass. On far m -scales, the residual exponent is

$$a_0 - \Re(B - C_1) - \frac{1}{2} \leq -\frac{c_u}{2} - \Re(B - C_1), \quad (6.26)$$

so the dyadic sum is uniform on the good patch. The estimates remain valid after A -derivatives, which only insert logarithms. This proves local normality and holomorphy. $\square$

We now isolate the compact-uniform statement actually needed when the variable regulators are removed. This is stronger than local uniform convergence of the assembled source and is therefore recorded separately. For $\epsilon \in \{d, q\}$ put

$$\begin{array}{c|ccc} \epsilon & B_\epsilon & C_\epsilon & \mathcal{X}_\epsilon(t) \\ \hline d & B & C_1 & 1 \\ q & -C_1 & -B & X_{B,C_1}(t). \end{array} \quad (6.27)$$

For $\rho \geq 1$, $d = (h, km)$, and $\mathfrak{r} = d\rho$, define

$$\begin{array}{c|cc} \sigma & y_\sigma(v) & x_\sigma(v) \\ + & v & v + \mathfrak{r} \\ - & v + \mathfrak{r} & v \end{array} \quad (v > 0), \quad l_\sigma(v) = \frac{y_\sigma(v)}{km}, \quad n_\sigma(v) = \frac{x_\sigma(v)}{h}. \quad (6.28)$$

Let $\mathcal{V}_\epsilon$ denote the complete coupled weight (3.14) with gamma ratio $g_{B_\epsilon, C_\epsilon}$. For fixed h, k set

$$\begin{aligned} & \mathfrak{i}_{\epsilon, \sigma}^{h, k}(A, B, C_1; m, \rho, v) \\ & := d h^{-1/2+C_\epsilon} k^{-1/2+A} \\ & \quad \times m^{-1+A-B_\epsilon} \\ & \quad \times \int_{\mathbb{R}} w(t) \mathcal{X}_\epsilon(t) y_\sigma(v)^{-1/2-A-it} \\ & \quad \times x_\sigma(v)^{-1/2-C_\epsilon+it} \mathcal{V}_\epsilon\left(\frac{y_\sigma(v)}{kmY}, \frac{mx_\sigma(v)}{h}, t\right) dt. \end{aligned} \quad (6.29)$$

Compact-uniform source pullback domination. Let

$$\mathcal{K}_A \subseteq \{\Re A < \tfrac{1}{2} - c_u/2\}, \quad \mathcal{K}_g \subseteq \Omega_g.$$

For fixed h, k , every branch (ϵ, σ) , and every fixed $j \geq 0$, there is a nonnegative function $D_{\epsilon, \sigma, j}^{h, k}(m, \rho, v)$ such that

$$\sup_{\mathcal{K}_A \times \mathcal{K}_g} \left| \partial_A^j \mathfrak{i}_{\epsilon, \sigma}^{h, k}(A, B, C_1; m, \rho, v) \right| \leq D_{\epsilon, \sigma, j}^{h, k}(m, \rho, v), \quad (6.30)$$

$$\sum_{m, \rho \geq 1} \int_0^\infty D_{\epsilon, \sigma, j}^{h, k}(m, \rho, v) dv < \infty. \quad (6.31)$$

The conclusion remains true after the finite coefficient-weighted sum over $h \leq H, k \leq K$.

Proof. Put

$$\begin{aligned} a_- &= \inf_{\mathcal{K}_A} \Re A, & a_+ &= \sup_{\mathcal{K}_A} \Re A, \\ b_- &= \inf_{\substack{\mathcal{K}_g \\ \text{epsilon} \in \{d, q\}}} \Re B_\epsilon, & b_+ &= \sup_{\substack{\mathcal{K}_g \\ \text{epsilon} \in \{d, q\}}} \Re B_\epsilon, \\ c_- &= \inf_{\substack{\mathcal{K}_g \\ \text{epsilon} \in \{d, q\}}} \Re C_\epsilon, & c_+ &= \sup_{\substack{\mathcal{K}_g \\ \text{epsilon} \in \{d, q\}}} \Re C_\epsilon. \end{aligned}$$

The two positive margins used below are

$$\delta_A = \frac{1}{2} - a_+ > \frac{c_u}{2}, \quad \delta_g = b_- - c_+ > 0. \quad (6.32)$$

The second inequality follows simultaneously for the direct and dual data from $\Re B > \kappa_0/\log T$ and $\Re C_1 < -\kappa_0/\log T$.

We use a branch-adapted disjoint partition. In both signs let

$$\mathcal{B}_\sigma = \{l_\sigma, n_\sigma \geq 1/2\}.$$

For $\sigma = +$ put

$$\mathcal{L}_+ = \{l_+ < 1/2\}, \quad \mathcal{N}_+ = \{l_+ \geq 1/2, n_+ < 1/2\},$$

and for $\sigma = -$ put

$$\mathcal{N}_- = \{n_- < 1/2\}, \quad \mathcal{L}_- = \{n_- \geq 1/2, l_- < 1/2\}.$$

These three sets are disjoint and cover the physical curve. Define the majorant in (6.30) explicitly by

$$D_{\epsilon, \sigma, j}^{h, k} = \sum_{\mathcal{R} \in \{\mathcal{B}_\sigma, \mathcal{N}_\sigma, \mathcal{L}_\sigma\}} \mathbf{1}_{\mathcal{R}} \sup_{\mathcal{K}_A \times \mathcal{K}_g} |\partial_A^j \mathfrak{I}_{\epsilon, \sigma}^{h, k}|. \quad (6.33)$$

We first put the parameter supremum inside one common pointwise estimate. For $r > 0$ and $\theta \in [\theta_-, \theta_+]$,

$$\sup_{\theta \in [\theta_-, \theta_+]} r^{-\theta} \leq r^{-\theta_-} + r^{-\theta_+}. \quad (6.34)$$

Since

$$dh^{-1/2+c} k^{-1/2+a} m^{-1+a-b} y^{-1/2-a} x^{-1/2-c} = \frac{d}{hk} m^{-3/2-b} l^{-1/2-a} n^{-1/2-c},$$

and, in the dual branch, compact-uniform Stirling gives

$$|\partial_t^r X_{B, C_1}(t)| \ll_{\mathcal{K}_{g, r}} T^{-r} T^{-\Re(B+C_1)} \ll_{\mathcal{K}_{g, r}} T^{-r},$$

the Fourier-conductor bound and (6.34) give, before any m, ρ, v integration or summation,

$$\begin{aligned} & \sup_{\mathcal{K}_A \times \mathcal{K}_g} |\partial_A^j \mathfrak{I}_{\epsilon, \sigma}^{h, k}| \\ & \ll_{\mathcal{K}_A, \mathcal{K}_g, j} T(\log T)^{C_{M, j}} \frac{d}{hk} m^{-3/2} (1 + |\log l|)^j (1 + l/Y)^{-P} (1 + mn/T)^{-Q} \\ & \quad \times (1 + \Delta |\log(x/y)|)^{-J} \sum_{\substack{a \in \{a_-, a_+\} \\ b \in \{b_-, b_+\} \\ c \in \{c_-, c_+\}}} m^{-b} l^{-1/2-a} n^{-1/2-c}. \end{aligned} \quad (6.35)$$

Here differentiation of the algebraic A -factors gives $\log(km/y) = -\log l$; derivatives of $\mathcal{G}$ insert only fixed polynomials in the complete contour variables and are absorbed by the same complete-contour bound. Thus only the eight displayed corners remain. It is now enough to prove integrability for each corner separately.

On $\mathcal{B}_\sigma$ use the product bound (3.15). Parameter supremums are absorbed by the piecewise powers

$$\omega_A(l) = \begin{cases} l^{-1/2-a_+}, & 1/2 \leq l \leq 1, \\ l^{-1/2-a_-}, & l > 1, \end{cases} \quad \omega_C(n) = \begin{cases} n^{-1/2-c_+}, & 1/2 \leq n \leq 1, \\ n^{-1/2-c_-}, & n > 1. \end{cases}$$

The half-mesh estimate (6.17), applied to the decreasing n -product tails on each of the two ranges separated at 1, gives, up to fixed h, k powers and a

logarithmic factor,

$$T \int_{1/2}^{\infty} \omega_A(l)(1+l/Y)^{-P} dl \sum_{m \geq 1} m^{-1/2-b_-} \times \left\{ \int_{1/2}^{\infty} \omega_C(n)(1+mn/T)^{-Q} dn + (1+m/T)^{-Q} \right\}. \quad (6.36)$$

For $m \leq T$, the integral term is $\ll_{\mathcal{H}_g} (T/m)^{1/2-c_-}$, and for $m > T$ the order Q may be chosen to give an arbitrary negative power. Consequently its residual m -sum is

$$\ll_{\mathcal{H}_g} T^{1/2-c_-} \sum_{m \leq T} m^{-1-(b_- - c_-)} \ll_{\mathcal{H}_g} T^{1/2-c_-},$$

while the mesh-supremum term is $O_{Q, \mathcal{H}_g}(T^{1/2-b_-})$ by (6.19). The l -integral is finite because its lower end is bounded away from zero and P is arbitrary at infinity. This proves (6.31) on the bulk.

We next treat the genuine n -crossing, which occurs for $\sigma = -$. Write $x = d\rho z$ and $X = md\rho/(hT)$. Uniformly for $c \in [c_-, c_+]$ and $0 < z \leq 1$, (6.21) gives

$$\int_0^1 z^{-1/2-c} (1+Xz)^{-Q} (1+|\log z|)^{-J} dz \ll_{\mathcal{H}_g} (1+X)^{-1/2+c} \{1+\log(2+X)\}^{-J}. \quad (6.37)$$

For $X > 1$, fix one of the eight corners and multiply (6.37) by the full algebraic prefactor before summing. The ρ -sum then cancels the m -dependent gcd $d = (h, km)$ exactly as in (6.22)–(6.23). Summing the finitely many corners gives

$$T^{1/2-c_-} Y^{1/2-a_-} \sum_{m \geq 1} m^{-1-\delta_g} < \infty. \quad (6.38)$$

For $X \leq 1$ one has $m\rho \leq hT/d$; hence there are only finitely many integer pairs, and (6.37) integrates the endpoint uniformly. On $z > 1$, the condition $n_- < 1/2$ gives $z < q/(2\rho)$ and $|\log(x/(x+d\rho))| \gg q^{-1} \geq H^{-1}$. In particular $\rho < q/2$ and $d\rho < x < h/2$, so the ρ - and v -ranges are finite for fixed h, k ; moreover $(1+mx/(hT))^{-Q} \leq (1+md/(hT))^{-Q}$ sums every remaining compact-dependent polynomial in m when Q is sufficiently large. Thus the whole $\mathcal{N}_-$ piece is integrable. In the noncrossing set $\mathcal{N}_+$, the inequalities $l_+ \geq 1/2$ and $n_+ < 1/2$ imply $km < h$ and $\rho < q/2$; only finitely many m, ρ occur, so the same product bounds suffice without a singular substitution.

For the genuine l -crossing, $\sigma = +$, put $y = d\rho z$. On $0 < z \leq 1$ the Fourier factor supplies $(1+|\log z|)^{-J}$ and

$$\int_0^1 \sup_{a \in [a_-, a_+]} z^{-1/2-a} (1+|\log z|)^{-J} dz = \int_0^1 z^{-1/2-a_+} (1+|\log z|)^{-J} dz < \infty \quad (6.39)$$

by $\delta_A > 0$. Dropping the additional long-variable decay and using the AFE conductor gives, for every corner and hence for their finite sum,

$$\sum_{\rho \geq 1} \rho^{-a-c} \left(1 + \frac{md\rho}{hT}\right)^{-Q} \ll_{\mathcal{K}_A, \mathcal{K}_g} \left(\frac{hT}{md}\right)^{1-a-c}, \quad (6.40)$$

$$\sum_{m \geq 1} m^{-2+2a-b+c} \leq \sum_{m \geq 1} m^{-1-2\delta_A-\delta_g} < \infty. \quad (6.41)$$

Here $a + c < 1$ follows from (6.32) and $c_+ < 0$. After the original prefactor is restored, the first line is explicitly

$$h^{1/2-a} k^{-1/2+a} T^{1-a-c} \sum_{m \geq 1} m^{-2+2a-b+c};$$

in particular the gcd d has cancelled before the m -sum is taken. For $z > 1$, dyadically sum $z \asymp Z$. The relations $d\rho Z \ll km$ and

$$\left| \log \frac{y}{y+d\rho} \right| \asymp Z^{-1}, \quad \frac{m(y+d\rho)}{hT} \gg \frac{md\rho Z}{hT},$$

give the two factors

$$(1 + \Delta/Z)^{-J}, \quad \left(1 + \frac{md\rho Z}{hT}\right)^{-Q}.$$

If $md\rho Z \leq hT$, then $m\rho Z \leq hT/d$, so only finitely many m , ρ , and dyadic Z occur and their total algebraic mass is a fixed compact-dependent polynomial. On the complement, choose Q larger than that polynomial exponent plus 4; the m -, ρ -, and Z -sums then converge absolutely. This proves integrability on $\mathcal{L}_+$. Finally, on the noncrossing set $\mathcal{L}_-$ one has $n_- \geq 1/2$ and $y_- = x + d\rho > 0$; there is no zero endpoint. The inequality $l_- < 1/2$ gives $d\rho < km/2$ and $h/2 \leq x < km/2$. Hence, for some fixed $C_{\mathcal{K}}$ containing the compact algebraic exponents, the absolute sum–integral is bounded by a constant times

$$\sum_{m \geq 1} m^{C_{\mathcal{K}}} \left(1 + \frac{m}{2T}\right)^{-Q},$$

after the polynomial number of ρ values and the v -interval have been included in $C_{\mathcal{K}}$. This converges when $Q > C_{\mathcal{K}} + 2$.

This proves (6.31) for $j = 0$. As recorded in (6.35), an A -derivative inserts a fixed polynomial in the complete contour variables and powers of $\log l$. For any $\eta > 0$, $|\log l|^j$ is bounded by $l^{-\eta}$ on $l \leq 1$ and by l^η on $l \geq 1$. Choose $\eta < \delta_A/4$ and then increase P, Q, J . All strict exponents above persist, proving the result for every fixed j . The sum over $h \leq H$, $k \leq K$ is finite; with $|a_h| \ll_\varepsilon h^\varepsilon$, $|b_k| \ll_\varepsilon k^\varepsilon$, the displayed estimates give a fixed polynomial in T, H, K , and the arbitrary Fourier/conductor orders absorb it wherever a power saving is required. This completes the proof. $\square$

6.3. Finite regulators and genuine Tonelli domains.

Proposition 6.4 (Four-branch regulator–Tonelli bridge). *Fix (B, C_1) in a compact subset of (6.15), keep $\Re u = c_u$, and put $\lambda = A + u$. Define*

$$z_d = \lambda + C_1 + s, \quad \nu_d = 1 - \lambda + B + s, \quad (6.42)$$

$$z_q = \lambda - B + s, \quad \nu_q = 1 - \lambda - C_1 + s. \quad (6.43)$$

Here d denotes the direct AFE piece and q the dual piece. In each of the four branches impose

$$\begin{aligned} m, |\rho| &\leq R, & R^{-1} &\leq l, n \leq R, \\ e^{-\kappa(m+|\rho|+l+n)}, & & R &\geq 2, \quad 0 < \kappa \leq 1. \end{aligned} \quad (6.44)$$

Then the source limits obtained in the order

$$V \longrightarrow \infty, \quad R \longrightarrow \infty, \quad \kappa \longrightarrow 0^+ \quad (6.45)$$

exist locally uniformly in the connected strip (6.16) and are holomorphic in the original shift A . The corresponding destination limits are absolutely convergent in the following four nonempty domains. The two negative branches use the common corridor

$$\frac{1}{2} < \Re \lambda < \frac{1}{2} + \Re(B - C_1). \quad (6.46)$$

For the direct positive and negative branches, respectively, the remaining conditions are

$$\Re \lambda < \frac{1}{2}, \quad (6.47)$$

$$\Re s > \max\{1 - \Re(\lambda + C_1), \Re(\lambda - B)\}.$$

$$\max\{1 - \Re(\lambda + C_1), \Re(\lambda - B)\} < \Re s < \frac{1}{2} - \Re C_1. \quad (6.48)$$

For the dual positive and negative branches, respectively, they are

$$\Re \lambda < \frac{1}{2}, \quad (6.49)$$

$$\Re s > \max\{1 - \Re(\lambda - B), \Re(\lambda + C_1)\}.$$

$$\max\{1 - \Re(\lambda - B), \Re(\lambda + C_1)\} < \Re s < \frac{1}{2} + \Re B. \quad (6.50)$$

For finite R, κ, V , the source and destination are the explicit objects defined in the proof and satisfy an exact rectangle identity, including its two horizontal s -edges. After $V \rightarrow \infty$ those edges vanish and the complete regulated source and destination agree exactly. After the remaining limits in (6.45), this equality extends from the relevant Tonelli open set to (6.16) by the one-variable identity theorem in A , with B, C_1 fixed. No continuation in the separate symbols $A + u$ and s is used.

Proof. We first define the two sides before taking any limit, using the branch data in (6.27) and the physical variables in (6.28). Thus $z_\epsilon = \lambda + C_\epsilon + s$ and $\nu_\epsilon = 1 - \lambda + B_\epsilon + s$. Let $[a]_V$ denote the upward segment $a - iV$ to $a + iV$, and set

$$E_{R,\kappa,\sigma}(m, \rho, v) = \mathbf{1}_{\{m, \rho \leq R\}} \mathbf{1}_{\{R^{-1} \leq l_\sigma(v), n_\sigma(v) \leq R\}} e^{-\kappa\{m+\rho+l_\sigma(v)+n_\sigma(v)\}}. \quad (6.51)$$

For $\lambda = A + u$, define

$$\begin{aligned} \mathfrak{J}_{\epsilon,\sigma}^{R,\kappa}(s, u; t, m, \rho) &:= d h^{-1/2+C_\epsilon+s} k^{-1/2+\lambda} Y^u \\ &\times m^{-1+\lambda-B_\epsilon-s} \int_0^\infty E_{R,\kappa,\sigma}(m, \rho, v) \\ &\times y_\sigma(v)^{-1/2-\lambda-it} x_\sigma(v)^{-1/2-C_\epsilon+it-s} dv. \end{aligned} \quad (6.52)$$

For fixed h, k , the finite source is

$$\begin{aligned} \mathcal{S}_{\epsilon,\sigma}^{R,\kappa,V}(A; h, k) &:= \int_{\mathbb{R}} w(t) \sum_{m \leq R} \sum_{\rho \leq R} \frac{1}{(2\pi i)^2} \int_{[c_u]_V} \int_{[2]_V} \Gamma(u) \frac{\mathcal{G}(s, u)}{s} \\ &\times \mathcal{X}_\epsilon(t) g_{B_\epsilon, C_\epsilon}(s, t) \mathfrak{J}_{\epsilon,\sigma}^{R,\kappa}(s, u; t, m, \rho) ds du dt. \end{aligned} \quad (6.53)$$

This is exactly the σ -branch of the endpoint-deleted zero mode after inserting the truncated coupled kernel (3.14); all sums and physical integrals are finite, so their order is immaterial. On a compact subset of the relevant Tonelli open set choose a single line $\Re s = \sigma_{\epsilon,\sigma} > 0$ satisfying the corresponding strict inequalities in (6.47)–(6.50). The finite destination is the same displayed expression with $[2]_V$ replaced by $[\sigma_{\epsilon,\sigma}]_V$:

$$\mathcal{S}_{\epsilon,\sigma}^{R,\kappa,V}(A; h, k) := \mathcal{S}_{\epsilon,\sigma}^{R,\kappa,V}(A; h, k) \big|_{[2]_V \mapsto [\sigma_{\epsilon,\sigma}]_V}. \quad (6.54)$$

No beta quotient or infinite arithmetic series is part of (6.54).

Write $\mathfrak{r} = d|\rho| > 0$. After the endpoint cutoffs have been deleted, the following four unregulated continuous integrals determine the Tonelli conditions. They are displayed separately because their zero endpoints differ; their beta evaluation is used only later, after all variable regulators have been removed.

For the direct, positive- ρ branch, $x = y + \mathfrak{r}$ and

$$\begin{aligned} I_{d,+}(\mathfrak{r}) &:= \int_0^\infty y^{-1/2-\lambda-it} (y + \mathfrak{r})^{-1/2-C_1+it-s} dy \\ &= \mathfrak{r}^{-z_d} \frac{\Gamma(\frac{1}{2} - \lambda - it) \Gamma(z_d)}{\Gamma(\frac{1}{2} + C_1 - it + s)}. \end{aligned} \quad (6.55)$$

At $y = 0$ this requires $\Re \lambda < 1/2$; at infinity it first requires $\Re z_d > 0$. The subsequent ρ -sum strengthens the latter to $\Re z_d > 1$. On that half-plane the gcd exponent $1 - z_d$ has negative real part, and the remaining m -sum is absolutely convergent when $\Re \nu_d > 1$. These conditions are exactly (6.47).

For the direct, negative- ρ branch, put $y = x + \mathfrak{r}$. Then

$$\begin{aligned} I_{d,-}(\mathfrak{r}) &:= \int_0^\infty (x + \mathfrak{r})^{-1/2-\lambda-it} x^{-1/2-C_1+it-s} dx \\ &= \mathfrak{r}^{-z_d} \frac{\Gamma(\frac{1}{2} - C_1 + it - s) \Gamma(z_d)}{\Gamma(\frac{1}{2} + \lambda + it)}. \end{aligned} \quad (6.56)$$

The endpoint $x = 0$ gives $\Re(C_1 + s) < 1/2$, while infinity, the ρ -sum, and the m -sum give respectively $\Re z_d > 0$, $\Re z_d > 1$, and $\Re \nu_d > 1$. Hence

$$\max\{1 - \Re(\lambda + C_1), \Re(\lambda - B)\} < \Re s < \frac{1}{2} - \Re C_1.$$

The first lower bound is below the upper bound precisely when $\Re \lambda > 1/2$; the second is below it precisely when $\Re \lambda < 1/2 + \Re(B - C_1)$. This proves (6.48), including the necessity of its corridor.

The dual AFE piece makes the substitution $(B, C_1) \mapsto (-C_1, -B)$ and has the fixed exterior factor $X_{B,C_1}(t)$. Its positive- ρ integral is, explicitly,

$$\begin{aligned} I_{q,+}(\mathfrak{r}) &:= \int_0^\infty y^{-1/2-\lambda-it} (y + \mathfrak{r})^{-1/2+B+it-s} dy \\ &= \mathfrak{r}^{-z_q} \frac{\Gamma(\frac{1}{2} - \lambda - it) \Gamma(z_q)}{\Gamma(\frac{1}{2} - B - it + s)}. \end{aligned} \quad (6.57)$$

The $y = 0$ condition is $\Re \lambda < 1/2$; the infinity, ρ -, and m -conditions are respectively $\Re z_q > 0$, $\Re z_q > 1$, and $\Re \nu_q > 1$. They give (6.49).

Finally, the dual, negative- ρ branch is

$$\begin{aligned} I_{q,-}(\mathfrak{r}) &:= \int_0^\infty (x + \mathfrak{r})^{-1/2-\lambda-it} x^{-1/2+B+it-s} dx \\ &= \mathfrak{r}^{-z_q} \frac{\Gamma(\frac{1}{2} + B + it - s) \Gamma(z_q)}{\Gamma(\frac{1}{2} + \lambda + it)}. \end{aligned} \quad (6.58)$$

Here $x = 0$ requires $\Re s < 1/2 + \Re B$; infinity and the two sums require $\Re z_q > 1$ and $\Re \nu_q > 1$. Therefore

$$\max\{1 - \Re(\lambda - B), \Re(\lambda + C_1)\} < \Re s < \frac{1}{2} + \Re B.$$

The two comparisons with the upper endpoint give separately $\Re \lambda > 1/2$ and $\Re \lambda < 1/2 + \Re(B - C_1)$, proving (6.50). Since $\Re(B - C_1) > 0$ on the good patch, both negative intervals are nonempty.

We next justify the contour operation before evaluating any beta integral. Keep R, κ fixed and truncate the complete s - and u -lines at a common height V . At this finite-regulator stage the possible singularities of the unreduced source Mellin integrand are as follows.

- (a) The pole $s = 0$ comes from $1/s$. Every source and destination s -line just used has positive real part, so it is not crossed.
- (b) The poles $u = 0, -1, -2, \dots$ come from $\Gamma(u)$. The line $\Re u = c_u > 0$ is not moved, so none is crossed.

(c) The numerator-gamma poles of g_{B,C_1} are

$$s = -\frac{1}{2} - B - it - 2j, \quad s = -\frac{1}{2} - C_1 + it - 2j \quad (j \geq 0),$$

and those of $g_{-C_1,-B}$ are

$$s = -\frac{1}{2} + C_1 - it - 2j, \quad s = -\frac{1}{2} + B + it - 2j \quad (j \geq 0).$$

For large T their real parts lie to the left of every positive line used above. Denominator gamma factors give zeros, not poles.

(d) The kernel $\mathcal{G}(s, u)$ is entire in (s, u) ; its displayed denominators depend only on the external shifts and stay bounded away from zero on the good patch. Because $m, |\rho|$ are finite at this stage, there is no arithmetic zeta function and hence no $z = 1$ or $\nu = 1$ pole.

Thus the finite rectangles deform without a residue. Let $\mathcal{I}_{\epsilon, \sigma}^{R, \kappa}(s, u, t)$ denote the finite m, ρ sum of the integrand in (6.53), with $w(t)$ and the differentials omitted. Cauchy's theorem gives the exact finite- V identity

$$\mathcal{J}_{\epsilon, \sigma}^{R, \kappa, V} - \mathcal{T}_{\epsilon, \sigma}^{R, \kappa, V} = \mathcal{H}_{\epsilon, \sigma}^{R, \kappa, V}, \quad (6.59)$$

where

$$\begin{aligned} \mathcal{H}_{\epsilon, \sigma}^{R, \kappa, V} = - \int_{\mathbb{R}} w(t) \frac{1}{(2\pi i)^2} \int_{[c_u]_V} \left\{ \int_{\sigma_{\epsilon, \sigma} - iV}^{2-iV} \right. \\ \left. + \int_{2+iV}^{\sigma_{\epsilon, \sigma} + iV} \right\} \mathcal{I}_{\epsilon, \sigma}^{R, \kappa}(s, u, t) ds du dt. \end{aligned} \quad (6.60)$$

The two horizontal integrals carry the displayed orientations. On either horizontal s -edge, $s = \sigma + iV$,

$$|e^{s^2}| = e^{\sigma^2 - V^2},$$

while the gamma ratios and remaining factors have at most exponential times polynomial growth in V . Hence $\mathcal{H}_{\epsilon, \sigma}^{R, \kappa, V} \rightarrow 0$ for fixed R, κ , locally uniformly in A . No u -contour is moved in this proposition. After the complete s -lines have been restored, the vertical u -tails are controlled by Stirling:

$$|\Gamma(c_u + iV)| \ll (1 + V)^{c_u - 1/2} e^{-\pi V/2}.$$

All other u -dependence is polynomial at fixed R, κ . Consequently, after $V \rightarrow \infty$ in (6.59), the two complete regulated vertical-contour integrals agree exactly. This is the first limit in (6.45); no variable regulator has yet been removed.

We now give the uniform dominators for the remaining two limits. Let

$$\mathcal{K}_A \subseteq \{\Re A < \tfrac{1}{2} - c_u/2\}, \quad \mathcal{K}_g \subseteq \Omega_g. \quad (6.61)$$

For fixed h, k and a branch (ϵ, σ) , remove $E_{R, \kappa, \sigma}$ from (6.52) and perform the complete t -, s -, and u -integrals on $\Re s = 2$, $\Re u = c_u$. Absolute convergence at fixed (m, ρ, v) and the definition (3.14) identify the result with the density

(6.29). Thus, after the already completed limit $V \rightarrow \infty$, the regulated source is exactly

$$\mathcal{S}_{\epsilon,\sigma}^{R,\kappa}(A; h, k) = \sum_{m,\rho \geq 1} \int_0^\infty E_{R,\kappa,\sigma}(m, \rho, v) i_{\epsilon,\sigma}^{h,k}(A, B, C_1; m, \rho, v) dv. \quad (6.62)$$

This is a pullback to the physical curve $(l, n) = (l_\sigma(v), n_\sigma(v))$; no supremum over an independent endpoint variable is taken.

Apply the compact-uniform source pullback statement proved above with $j = 0$ and write its explicit three-piece majorant (6.33) as $D_{\epsilon,\sigma,0}^{h,k}$. Then

$$\sum_{m,\rho \geq 1} \int_0^\infty D_{\epsilon,\sigma,0}^{h,k}(m, \rho, v) dv < \infty. \quad (6.63)$$

Since $0 \leq E_{R,\kappa,\sigma} \leq 1$, the summable pullback majorant (6.63) is independent of R and κ . For every fixed (m, ρ, v) the factor in (6.51) tends first, as $R \rightarrow \infty$, to $e^{-\kappa(m+\rho+l_\sigma(v)+n_\sigma(v))}$ and then, as $\kappa \rightarrow 0^+$, to 1. Dominated convergence in the measure $\sum_{m,\rho} dv$, applied also to the A -derivatives, now proves the two source limits in the asserted order and locally uniformly on the stated A -compact sets.

For a destination, restrict temporarily to a compact subset $\mathcal{K}_{\text{Ton}}$ of its branchwise Tonelli open set, and choose the line $\Re s = \sigma_{\epsilon,\sigma}$ with a common margin $\delta > 0$ such that

$$\Re z_\epsilon \geq 1 + \delta, \quad \Re \nu_\epsilon \geq 1 + \delta, \quad (6.64)$$

and the relevant zero-endpoint inequality in (6.47)–(6.50) also has margin δ . After scaling v by $\mathfrak{r} = d\rho$, define

$$\begin{aligned} b_{\epsilon,+}(v) &= \sup_{\mathcal{K}_{\text{Ton}}} v^{-1/2-\Re \lambda} (1+v)^{-1/2-\Re C_\epsilon - \Re s}, \\ b_{\epsilon,-}(v) &= \sup_{\mathcal{K}_{\text{Ton}}} v^{-1/2-\Re C_\epsilon - \Re s} (1+v)^{-1/2-\Re \lambda}. \end{aligned}$$

The endpoint margins give $\int_0^\infty b_{\epsilon,\sigma}(v) dv < \infty$. Stirling and the Gaussian factor in $\mathcal{G}$ give, on the complete contours,

$$C_{\mathcal{K}_{\text{Ton}}} (1 + |\tau_u| + |\tau_s|)^N e^{-\pi|\tau_u|/3 - \tau_s^2/2} m^{-1-\delta} \rho^{-1-\delta} b_{\epsilon,\sigma}(v), \quad (6.65)$$

for some fixed N . Indeed the gcd factor after scaling is $d^{1-\Re z_\epsilon} \leq 1$. Thus (6.65) is integrable in $(\tau_u, \tau_s, m, \rho, v)$ and is independent of R, κ . It justifies the same ordered limits at every destination, uniformly on $\mathcal{K}_{\text{Ton}}$. Notice that destination domination is asserted only inside a Tonelli chamber; outside it, holomorphy comes from the source and the identity theorem, not from an unproved absolute majorant.

The recovery order is now strict. First restore the complete contours ($V \rightarrow \infty$), then remove the geometric truncation ($R \rightarrow \infty$) while $\kappa > 0$, and then remove the Abel factor ($\kappa \rightarrow 0^+$). Only after the last step are the unregulated beta integrals (6.55)–(6.58) evaluated. Their powers $\mathfrak{r}^{-z_\epsilon} = d^{-z_\epsilon} \rho^{-z_\epsilon}$ then permit, by absolute convergence in the Tonelli chamber, first the ρ -sum $\sum_{\rho \geq 1} \rho^{-z_\epsilon} = \zeta(z_\epsilon)$ and then the m - and gcd-sum giving $C_{h,k}$.

The zeta functional equation and the finite Euler identity of Lemma 7.1 are applied only after these series have been recovered. Thus no Abel-weighted integral is identified with a beta quotient, and no Abel-weighted arithmetic sum is identified with an Euler product.

For completeness, singularities appearing only after the variable regulators have been removed must be recorded separately. The direct branches produce $\zeta(z_d)$ and a gcd- m factor with polar part $\zeta(\nu_d)$; the dual branches similarly produce $\zeta(z_q)$ and $\zeta(\nu_q)$.

- (i) The direct pole $z_d = 1$ is cancelled by the kernel zero $s = 1 - \lambda - C_1$, and the dual pole $z_q = 1$ by the zero $s = 1 - \lambda + B$.
- (ii) After applying the zeta functional equation, the direct hyperplanes $z_d = 0$, $\nu_d = 1$, and the corresponding dual hyperplanes $z_q = 0$, $\nu_q = 1$ remain. They are not discarded or crossed branchwise: these are the small arithmetic divisors cancelled only after the four branches and both diagonals are assembled.
- (iii) Before the positive- and negative- ρ pieces are combined, the beta factors may have the following gamma poles, where $j \geq 0$:

$$\begin{aligned} (d, +) : \quad z_d &= -j, \quad \lambda = \frac{1}{2} + j - it; \\ (d, -) : \quad z_d &= -j, \quad s = \frac{1}{2} - C_1 + it + j; \\ (q, +) : \quad z_q &= -j, \quad \lambda = \frac{1}{2} + j - it; \\ (q, -) : \quad z_q &= -j, \quad s = \frac{1}{2} + B + it + j. \end{aligned}$$

They lie outside the four Tonelli domains. After combining the two ρ -signs, Lemma 7.2 cancels every apparent near-real archimedean pole. The remaining direct poles are

$$s = n - \frac{1}{2} - C_1 + it \quad (n \geq 1), \quad s = -\frac{1}{2} - B - it - 2j, \quad s = -\frac{1}{2} - C_1 + it - 2j,$$

and the remaining dual poles are

$$s = n - \frac{1}{2} + B + it \quad (n \geq 1), \quad s = -\frac{1}{2} + C_1 - it - 2j, \quad s = -\frac{1}{2} + B + it - 2j.$$

All have imaginary height comparable with T . If a later fixed-line deformation crosses one, e^{s^2} makes its residue smaller than every power of T . Reciprocal gamma factors again give zeros only.

The poles $s = 0$ and those of $\Gamma(u)$ have not moved during this bridge; they are treated in the subsequent fixed- u assembly and final u -move.

It remains to identify the selected continuation. The positive branches agree with their destinations on the nonempty open set

$$\mathcal{U}_+ = \{A : \Re A + c_u < 1/2\} \cap \{\Re A < 1/2 - c_u/2\},$$

whereas the negative branches agree on

$$\mathcal{U}_- = \left\{ A : \frac{1}{2} < \Re A + c_u < \frac{1}{2} + \Re(B - C_1) \right\} \cap \{ \Re A < 1/2 - c_u/2 \}.$$

The latter is nonempty because $\Re(B - C_1) > 0$; for large T its width is $O(1/\log T) < c_u/2$, so it lies in the connected strip (6.16). Apply the one-variable identity theorem separately to each complete, already integrated branch as a function of A . This extends an exact equality to the target A ; it does not propagate the endpoint-deletion $O(T^{-N})$ error and does not treat λ and s as independent continuation variables. $\square$

7. BETA INTEGRALS, EULER FACTORS, AND GAMMA IDENTITIES

We now evaluate the endpoint-deleted zero mode at the Tonelli destinations constructed in Section 6. Put

$$\lambda = \alpha + u, \quad z = \lambda + \gamma + s, \quad q_1 = 1 - z, \quad v_1 = 1 - \lambda + \beta + s. \quad (7.1)$$

Every expression in this section is the fully unregulated destination limit of Proposition 6.4: the complete s, u contours have already been restored, R has gone to infinity, and the Abel parameter κ has already gone to zero. We first evaluate the ordinary beta integral, then recover the absolutely convergent ρ -series, and only then recover the m - and gcd-series. Thus none of the beta or Euler identities below is applied to an Abel-weighted expression.

For $r > 0$, ordinary beta integration gives

$$\begin{aligned} & \int_0^\infty y^{-1/2-\lambda-it} (y+r)^{-1/2-\gamma+it-s} dy \\ &= r^{-z} \frac{\Gamma(\frac{1}{2} - \lambda - it) \Gamma(z)}{\Gamma(\frac{1}{2} + \gamma - it + s)}. \end{aligned} \quad (7.2)$$

A genuine absolute-convergence domain, including the subsequent ρ - and m -sums, is

$$\Re \lambda < \frac{1}{2}, \quad \Re z > 1, \quad \Re v_1 > 1. \quad (7.3)$$

For $r = -R < 0$, substitute $y = R + x$ to obtain

$$\begin{aligned} & \int_0^\infty (x+R)^{-1/2-\lambda-it} x^{-1/2-\gamma+it-s} dx \\ &= R^{-z} \frac{\Gamma(\frac{1}{2} - \gamma + it - s) \Gamma(z)}{\Gamma(\frac{1}{2} + \lambda + it)}. \end{aligned} \quad (7.4)$$

Here a genuine domain is

$$\Re(\gamma + s) < \frac{1}{2}, \quad \Re z > 1, \quad \Re v_1 > 1. \quad (7.5)$$

The two signs need not have a common elementary chamber near the intended shifts; Section 6 proves the two identities separately and then selects their common meromorphic continuation by regulated exact equality.

The sum over ρ supplies $\zeta(z)$. We use

$$\zeta(z) = \chi(z)\zeta(1-z), \quad \chi(z) = \pi^{z-1/2} \frac{\Gamma((1-z)/2)}{\Gamma(z/2)}. \quad (7.6)$$

The remaining gcd sum is

$$C_{h,k}(s, u) = \sum_{m \geq 1} \frac{(h, km)^{q_1}}{m^{v_1}}. \quad (7.7)$$

Lemma 7.1 (Finite Euler factors). *In the common meromorphic continuation,*

$$h^{-1/2+\gamma+s} k^{-1/2+\lambda} \zeta(1-z) C_{h,k}(s, u) = Z_{-\gamma-s, \beta+s, -\lambda}(h, k). \quad (7.8)$$

Proof. Write $g = (h, k)$, $h = gh_0$, $k = gk_0$. Since $(h, km) = g(h_0, m)$,

$$C_{h,k} = g^{q_1} \zeta(v_1) \prod_{p^\nu \parallel h_0} (1 - p^{-v_1}) \sum_{j \geq 0} p^{q_1 \min(j, \nu) - v_1 j}. \quad (7.9)$$

For $X = p^{-q_1}$ and $Y_0 = p^{-v_1}$, splitting the sum at $j = \nu$ gives

$$(1 - Y_0) \sum_{j \geq 0} X^{-\min(j, \nu)} Y_0^j = X^{-\nu} (1 - X) (1 - Y_0) \sum_{i+j \geq \nu} X^i Y_0^j. \quad (7.10)$$

This is exactly the local factor in (2.5), after the exterior powers in (7.8) are restored. $\square$

Combine the two signs in the archimedean factor

$$\begin{aligned} \Phi_{\lambda; \beta, \gamma}(s, t) = \chi(z) \Gamma(z) g_{\beta, \gamma}(s, t) & \left\{ \frac{\Gamma(\frac{1}{2} - \lambda - it)}{\Gamma(\frac{1}{2} + \gamma - it + s)} \right. \\ & \left. + \frac{\Gamma(\frac{1}{2} - \gamma + it - s)}{\Gamma(\frac{1}{2} + \lambda + it)} \right\}. \end{aligned} \quad (7.11)$$

There is no additional π^{-s} in (7.11), because it is already part of $g_{\beta, \gamma}$.

Lemma 7.2 (Archimedean pole collapse). *Let $a_0 = \frac{1}{2} - \lambda - it$. Then*

$$\begin{aligned} \chi(z) \Gamma(z) & \left\{ \frac{\Gamma(a_0)}{\Gamma(a_0 + z)} + \frac{\Gamma(1 - a_0 - z)}{\Gamma(1 - a_0)} \right\} \\ & = (2\pi)^z \frac{\Gamma(a_0)}{\Gamma(a_0 + z)} \frac{\sin \pi(a_0 + z/2)}{\sin \pi(a_0 + z)}. \end{aligned} \quad (7.12)$$

For $t \asymp T$ and $|\Im \lambda| + |\Im s| \leq (\log T)^2$, the left side has no pole near the real s -axis. Its possible poles occur at

$$s = n - \frac{1}{2} - \gamma + it \quad (n \geq 1), \quad (7.13)$$

and the numerator gamma factors of g add only

$$s = -\frac{1}{2} - \beta - it - 2j, \quad s = -\frac{1}{2} - \gamma + it - 2j \quad (j \geq 0). \quad (7.14)$$

All are at imaginary height comparable with T .

Proof. Duplication and reflection give

$$\chi(z)\Gamma(z) = \frac{(2\pi)^z}{2\cos(\pi z/2)}$$

and

$$\frac{\Gamma(1-a_0-z)}{\Gamma(1-a_0)} = \frac{\Gamma(a_0)}{\Gamma(a_0+z)} \frac{\sin \pi a_0}{\sin \pi(a_0+z)}.$$

Now use

$$\sin \pi(a_0+z) + \sin \pi a_0 = 2 \sin \pi(a_0+z/2) \cos(\pi z/2).$$

This proves (7.12), including cancellation of the apparent odd-integer poles of $1/\cos(\pi z/2)$. At nonpositive integral values of a_0+z , the reciprocal gamma zero cancels the sine denominator. The remaining poles are (7.13) and (7.14). If such a pole is crossed between fixed vertical lines, the factor e^{s^2} makes its residue $O_D(T^{-D})$. $\square$

Lemma 7.3 (Archimedean reflection). *As meromorphic identities,*

$$\Phi_{\lambda;\beta,\gamma}(0,t) = X_{\lambda,\gamma}(t), \quad (7.15)$$

and

$$X_{\beta,\gamma}(t)\Phi_{\lambda;-\gamma,-\beta}(s,t) = \Phi_{\lambda;\beta,\gamma}(-s,t). \quad (7.16)$$

Proof. Substitute (2.6), (7.6), and (7.11). Pair the full gamma factors by

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$$

and the half-arguments by

$$\Gamma(z)\Gamma(z+\frac{1}{2}) = 2^{1-2z}\sqrt{\pi}\Gamma(2z).$$

For (7.15), the two terms in braces combine by reflection and the remaining half-gamma factors form $X_{\lambda,\gamma}$. For (7.16), first combine the β, γ factors with $X_{\beta,\gamma}$ and then apply duplication. The identities hold on a pole-free open set and hence everywhere by meromorphic continuation. $\square$

We next record the contour step that turns the four branchwise destinations into the common contours used in the residue assembly.

Lemma 7.4 (Common-contour recovery). *Fix $\mathcal{K}_g \Subset \Omega_g$ and a compact set of target values $\mathcal{K}_A \Subset \{A : \Re A < 1/2 - c_u/2\}$. Assume, throughout $\mathcal{K}_A \times \mathcal{K}_g$, that*

$$\max\{|\Re(A+C_1)|, |\Re(A-B)|\} \leq \frac{C_*}{\log T}. \quad (7.17)$$

Suppose

$$c_u + \frac{C_*}{\log T} < c_s < \min\left\{\frac{1}{2}, 1 - c_u - \frac{C_*}{\log T}\right\}. \quad (7.18)$$

After the limits in (6.45), pair the positive- and negative- ρ destinations within each AFE sign. Their continuations selected by Proposition 6.4 are represented on the single line $\Re s = c_s$ by (7.20) and (7.21) below, with the explicit remote-pole remainders recorded there. For every fixed N their coefficient-weighted h, k -sum is $O_N(T^{-N})$, locally uniformly on $\mathcal{K}_A \times \mathcal{K}_g$.

Proof. Start each branch in its own Tonelli chamber and on the line $\Re s = \sigma_{\epsilon, \sigma}$ used in (6.54). The four starting values of A and their paths to a given target in $\mathcal{K}_A$ are chosen branchwise; in particular, no common Tonelli starting point for the positive and negative branches is asserted. Along each path continue only the external variable A in (6.16), as in the last paragraph of the proof of Proposition 6.4, and simultaneously deform the s -line continuously to $\Re s = c_s$. The line can be chosen throughout the path so that, with a positive margin uniform on $\mathcal{K}_A \times \mathcal{K}_g$,

$$\begin{aligned} \Re(-\lambda - C_1), \Re(\lambda - B) &< \Re s && \text{in the direct branches,} \\ \Re(-\lambda + B), \Re(\lambda + C_1) &< \Re s && \text{in the dual branches.} \end{aligned}$$

At the Tonelli endpoint these inequalities follow from (6.47)–(6.50); at the target they follow from (7.17) and (7.18). A continuous choice is obtained, separately for each branch, by taking a continuous strict majorant of 0 and that branch's two displayed affine real parts, and then smoothing it inside the open margin. Thus the arithmetic divisors $z_d = 0$, $\nu_d = 1$, $z_q = 0$, and $\nu_q = 1$ remain strictly to the left of the moving line. The pole $s = 0$ does also. The potential poles $z_\epsilon = 1$ are cancelled identically by the corresponding zeros of $\mathcal{G}(s, u)$.

We spell out the complete-contour bound without asserting domination through a pole crossing. First truncate both contours and perturb each branchwise path and moving line into general position. There are then only finitely many remote crossings, and the deformation splits into finitely many closed pole-free subpaths. On each such subpath the integrand stays a positive distance from its polar divisors. For $|\Im u| + |\Im s| \leq (\log T)^2$, uniform Stirling estimates give a fixed polynomial majorant. On the complementary part, e^{s^2} is Gaussian in $\Im s$, while $\Gamma(c_u + i\tau_u)$ and the gamma quotients give an integrable exponential times a polynomial in $|\tau_u| + |\tau_s| + T$. These bounds are uniform on the given pole-free subpath and after the finite h, k -sum. In a small neighbourhood of a remote crossing the contour is indented and the difference is recorded as its residue; no dominating function across the singular parameter value is used. This justifies Cauchy's theorem on every truncated pole-free piece and then removal of the truncations. The final s -line is itself pole-free: on the good patch the nearest pole in (7.13) has real part $1/2 - \Re C_\epsilon > 1/2$, the poles in (7.14) lie to the left, and (7.18) has $c_s < 1/2$. The final u -line is also pole-free by (7.17) and $c_u < 1/4$.

After the two ρ -signs are paired, Lemma 7.2 leaves only the poles (7.13)–(7.14); if the deformation crosses one, its imaginary part is comparable with $t \asymp T$ and the factor e^{s^2} makes its total residue $O_N(T^{-N})$ for every fixed N . Moving A may also move the numerator-gamma poles

$$u = \frac{1}{2} + j - A - it \quad (j \geq 0) \quad (7.19)$$

through $\Re u = c_u$. Only finitely many j can be crossed on the compact deformation. At $u_j = 1/2 + j - A - it$ one has $a_0 = -j$. Away from a

simultaneous s -pole, the residue of the collapsed factor in (7.12) is, up to a sign,

$$\frac{(2\pi)^{z_j}}{2j! \Gamma(z_j - j) \cos(\pi z_j/2)}, \quad z_j = \frac{1}{2} + j + C_\epsilon + s - it,$$

where $C_d = C_1$ and $C_q = -B$. Stirling shows that the exponential parts of the reciprocal gamma and cosine factors cancel, leaving polynomial growth on a fixed s -line. On the other hand,

$$|\Gamma(u_j)Y^{u_j}| \ll_{j, \mathcal{K}_A} Y^{1/2+j-\Re A} (1+T)^{j-\Re A} e^{-\pi T/16}.$$

All remaining factors and the finite coefficient-weighted h, k -sum have at most polynomial growth, while the s -integral retains its Gaussian. Consequently every crossed u -residue is $O_N(T^{-N})$, uniformly on the stated compact sets. Likewise a crossed pole from (7.13)–(7.14) has $|e^{s_*^2}| \leq e^{-T^2/50}$ for all sufficiently large T ; its other factors have at most exponential-linear times polynomial growth, so its total residue is $O_N(T^{-N})$.

For clarity about possible multiplicities, use the general-position perturbation just introduced. The positive margin above keeps 0 and all small arithmetic divisors on the same side, while remote s - and u -divisors are crossed one at a time. Apply the one-variable residue theorem successively and then let the perturbation tend to zero. If remote divisors meet in the limit, the resulting term is the corresponding iterated residue. A mixed s - u residue retains both the $e^{-T^2/50}$ Gaussian from the remote s -value and the $e^{-\pi T/16}$ factor from $\Gamma(u_j)$; a repeated pole only inserts a fixed polynomial factor. Hence all such terms are still $O_N(T^{-N})$. Denominator gamma factors contribute zeros and can only lower these multiplicities. Removing the contour truncations is justified by the complete-contour majorant above. These s - and u -residues, including the iterated ones, are exactly the remote-pole remainders in the statement. No small arithmetic residue is crossed or discarded.

This moving-line argument represents the one-variable continuation in A already selected by Proposition 6.4; it is not a separate continuation in λ and s . Finally, adding the two beta quotients in each AFE sign gives (7.11), while Lemma 7.1 gives the displayed Z -factor. This proves the two common-contour formulas. $\square$

The endpoint-deleted direct and dual zero modes are therefore

$$\begin{aligned} \tilde{\mathcal{O}}_{0,Y}^+(h, k) &= \int_{\mathbb{R}} w(t) \frac{1}{2\pi i} \int_{(c_u)} \Gamma(u) Y^u \frac{1}{2\pi i} \int_{(c_s)} \frac{\mathcal{G}(s, u)}{s} \Phi_{\lambda; \beta, \gamma}(s, t) \\ &\quad \times Z_{-\gamma-s, \beta+s, -\lambda}(h, k) ds du dt \\ &\quad + \mathcal{E}_{\text{rem}}^+(h, k), \\ \tilde{\mathcal{O}}_{0,Y}^-(h, k) &= \int_{\mathbb{R}} w(t) \frac{1}{2\pi i} \int_{(c_u)} \Gamma(u) Y^u \frac{1}{2\pi i} \int_{(c_s)} \frac{\mathcal{G}(s, u)}{s} X_{\beta, \gamma}(t) \Phi_{\lambda; -\gamma, -\beta}(s, t) \\ &\quad \times Z_{\beta-s, -\gamma+s, -\lambda}(h, k) ds du dt \end{aligned} \tag{7.20}$$

$$+ \mathcal{E}_{\text{rem}}^-(h, k). \quad (7.21)$$

For every fixed N , locally uniformly on the compact shift sets in Lemma 7.4,

$$\sum_{h \leq H} \sum_{k \leq K} \frac{|a_h b_k|}{\sqrt{hk}} (|\mathcal{E}_{\text{rem}}^+(h, k)| + |\mathcal{E}_{\text{rem}}^-(h, k)|) \ll_N T^{-N}. \quad (7.22)$$

The branchwise beta and Euler calculations are made before this common line is introduced. In particular, neither formula is obtained by moving a negative branch at fixed A across a small arithmetic divisor.

8. FIXED- u RESIDUE ASSEMBLY AND QUANTITATIVE CONTINUATION

For $\lambda = \alpha + u$, define the complete three-permutation expression

$$\begin{aligned} \mathcal{P}_\lambda(h, k; t) &= Z_{\lambda, \beta, \gamma}(h, k) + X_{\lambda, \gamma}(t) Z_{-\gamma, \beta, -\lambda}(h, k) \\ &\quad + X_{\beta, \gamma}(t) Z_{\lambda, -\gamma, -\beta}(h, k). \end{aligned} \quad (8.1)$$

The individual terms are meromorphic. They will always be combined before a collision divisor is crossed.

The direct and dual diagonal terms for fixed h, k are

$$\begin{aligned} \mathcal{D}^+(h, k) &= \int_{\mathbb{R}} w(t) \frac{1}{2\pi i} \int_{(c_u)} \Gamma(u) Y^u \\ &\quad \times \frac{1}{2\pi i} \int_{(c_s)} \frac{\mathcal{G}(s, u)}{s} g_{\beta, \gamma}(s, t) Z_{\lambda, \beta+s, \gamma+s}(h, k) ds du dt, \quad (8.2) \\ \mathcal{D}^-(h, k) &= \int_{\mathbb{R}} w(t) \frac{1}{2\pi i} \int_{(c_u)} \Gamma(u) Y^u X_{\beta, \gamma}(t) \\ &\quad \times \frac{1}{2\pi i} \int_{(c_s)} \frac{\mathcal{G}(s, u)}{s} g_{-\gamma, -\beta}(s, t) Z_{\lambda, -\gamma+s, -\beta+s}(h, k) ds du dt. \end{aligned} \quad (8.3)$$

8.1. The three residue cancellations. Reflect the dual diagonal and dual zero mode by $s \mapsto -s$. Before any term is separated, the small arithmetic poles at fixed u are

term	pole locations before reflection where indicated
$\mathcal{D}^+$	$s = -\lambda - \gamma, s = -(\beta + \gamma)/2$
$\mathcal{D}^-$	$s = \beta - \lambda, s = (\beta + \gamma)/2$
zero mode	$s = -\lambda - \gamma, s = \lambda - \beta$

The exact identities required at these poles are

$$X_{\beta, \gamma}(t) g_{-\gamma, -\beta}(-s, t) = g_{\beta, \gamma}(s, t) X_{\beta+s, \gamma+s}(t), \quad (8.4)$$

$$\Phi_{\lambda; \beta, \gamma}(-\lambda - \gamma, t) = g_{\beta, \gamma}(-\lambda - \gamma, t), \quad (8.5)$$

$$\Phi_{\lambda; \beta, \gamma}(\lambda - \beta, t) = g_{\beta, \gamma}(\lambda - \beta, t) X_{\lambda, \gamma+\lambda-\beta}(t). \quad (8.6)$$

They follow from (2.6), (7.11), reflection and duplication.

At $s = -\lambda - \gamma$, the direct diagonal and zero-mode arithmetic factors both equal

$$Z_{\lambda, \beta - \lambda - \gamma, -\lambda}(h, k), \quad (8.7)$$

while their polar zeta arguments have derivatives $+1$ and -1 . Equation (8.5) therefore cancels the residues. After reflection, the second pair at $s = \lambda - \beta$ has common factor

$$Z_{-\gamma - \lambda + \beta, \lambda, -\lambda}(h, k), \quad (8.8)$$

and (8.6) gives opposite residues. Finally, at $s_0 = -(\beta + \gamma)/2$, the two diagonal factors specialize to

$$Z_{\lambda, (\beta - \gamma)/2, (\gamma - \beta)/2}(h, k). \quad (8.9)$$

Their polar arguments have derivatives $+2$ and -2 , and $X_{(\beta - \gamma)/2, (\gamma - \beta)/2}(t) = 1$. Thus the third pair also cancels. The equalities in (8.7)–(8.9) are equalities of the complete Euler products, not just their global zeta factors.

Choose

$$\sigma_s = -\frac{1}{2} + \varepsilon_s, \quad 0 < \varepsilon_s < \min\left(\frac{1}{4}, \frac{1}{2} - c_u\right), \quad (8.10)$$

and choose C_* so that every signed pair of real shift parts occurring below has absolute value at most $C_*/\log T$. For large T , take

$$c_u + \frac{C_*}{\log T} < c_s < \min\left(\frac{1}{2} - \varepsilon_s, 1 - c_u - \frac{C_*}{\log T}\right). \quad (8.11)$$

In the central strip $-c_s < \Re s < c_s$, the complete reflected integrand has only

$$\begin{array}{l|l} \begin{array}{l} x = s + \lambda + \gamma = 0 \\ y = s - \lambda + \beta = 0 \\ z = 2s + \beta + \gamma = x + y = 0 \\ s = 0 \end{array} & \begin{array}{l} s = -\lambda - \gamma, \\ s = \lambda - \beta, \\ s = -(\beta + \gamma)/2, \\ s = 0. \end{array} \end{array} \quad (8.12)$$

The $z = 1$ locations lie outside by (8.11) and are zeros of $\mathcal{G}$ in any wider deformation. By Lemma 7.2, all other archimedean poles have imaginary part comparable with t and are Gaussian-small.

By Lemma 7.4, the four branchwise Tonelli destinations may now be paired and represented on this common c_s -line. This conclusion was obtained by moving A and the branch line together, with every small arithmetic divisor kept on the same side; it is not a branchwise fixed- A contour shift. Consequently the following contour difference represents the continuation selected in Proposition 6.4, up to an error smaller than every power of T after the outer sum.

Put

$$F_0(s) = \mathcal{G}(s, u) \Phi_{\lambda; \beta, \gamma}(s, t) Z_{-\gamma - s, \beta + s, -\lambda}(h, k). \quad (8.13)$$

Using (7.16), evenness of $\mathcal{G}$, and symmetry of Z in its first two shifts, the dual zero mode becomes exactly

$$-\frac{1}{2\pi i} \int_{(-c_s)} \frac{F_0(s)}{s} ds. \quad (8.14)$$

The sign comes from $1/(-s)$; reversal of the parametrized contour introduces no second sign. Thus the two zero modes form a contour difference, evaluated by the $x = 0$, $y = 0$, and $s = 0$ residues, with no zero-mode boundary line.

Move the direct diagonal from (c_s) to (σ_s) and the reflected dual diagonal from $(-c_s)$ to $(-\sigma_s)$. The three nonzero residue pairs cancel as above. The $s = 0$ residues of the two diagonals and the zero mode are precisely the three terms in $Q(u)\mathcal{P}_\lambda$. Undoing the reflection on the final dual line leaves the full diagonal boundary

$$\begin{aligned} \mathcal{R}_{\text{diag},Y}(h,k) = & \int_{\mathbb{R}} w(t) \frac{1}{2\pi i} \int_{(c_u)} \Gamma(u) Y^u \\ & \times \left\{ \frac{1}{2\pi i} \int_{(\sigma_s)} \frac{\mathcal{G}(s,u)}{s} g_{\beta,\gamma}(s,t) \right. \\ & \quad \times Z_{\alpha+u,\beta+s,\gamma+s}(h,k) ds \\ & + X_{\beta,\gamma}(t) \frac{1}{2\pi i} \int_{(\sigma_s)} \frac{\mathcal{G}(s,u)}{s} g_{-\gamma,-\beta}(s,t) \\ & \quad \times Z_{\alpha+u,-\gamma+s,-\beta+s}(h,k) ds \left. \right\} du dt. \end{aligned} \quad (8.15)$$

The two terms in (8.15) are separated only after the joint cancellation at the three small divisors.

Combining Proposition 6.2 with this fixed- u residue calculation yields, on the good patch and after the outer coefficient sum,

$$\begin{aligned} \mathcal{D}^+ + \mathcal{D}^- + \mathcal{O}_{0,Y}^+ + \mathcal{O}_{0,Y}^- \\ = \int_{\mathbb{R}} w(t) \frac{1}{2\pi i} \int_{(c_u)} \Gamma(u) Y^u Q(u) \mathcal{P}_{\alpha+u}(h,k;t) du dt \\ + \mathcal{R}_{\text{diag},Y}(h,k) + O_N(T^{-N}) \end{aligned} \quad (8.16)$$

for every $N > 0$. This error includes, in particular, the coefficient-weighted contribution of $\mathcal{E}_{\text{rem}}^+(h,k) + \mathcal{E}_{\text{rem}}^-(h,k)$ from (7.20)–(7.21), bounded by (7.22). As usual in an h,k identity, the error in (8.16) is understood after summation with $a_h \bar{b}_k / \sqrt{hk}$.

External-shift objects. For external shifts A, B, C_1 , let $\mathcal{A}_T(A, B, C_1)$ denote the complete source: both diagonals and both exact zero modes, with their endpoint cutoffs, common positive u -line, and outer coefficient sum. Let

$$\begin{aligned} \mathcal{R}_T(A, B, C_1) = & \sum_{h \leq H} \sum_{k \leq K} \frac{a_h \bar{b}_k}{\sqrt{hk}} \\ & \times \int_{\mathbb{R}} w(t) \frac{1}{2\pi i} \int_{(c_u)} \Gamma(u) Y^u Q(u) \mathcal{P}_{A+u}(h,k;t) du dt, \end{aligned} \quad (8.17)$$

$$\mathcal{D}_T(A, B, C_1) = \sum_{h \leq H} \sum_{k \leq K} \frac{a_h \bar{b}_k}{\sqrt{hk}} \mathcal{R}_{\text{diag}, Y}(h, k). \quad (8.18)$$

The shifts in every displayed kernel are changed from (α, β, γ) to (A, B, C_1) .

Lemma 8.1 (Rapid-decay propagation). *Let $\Omega \subset \mathbb{C}^n$ be a fixed connected domain, let $U \Subset \Omega$ be nonempty and open, and let $K_0 \Subset \Omega$. If f_T is holomorphic on Ω and*

$$\sup_{\Omega} |f_T| \leq T^{C_2}, \quad \sup_U |f_T| \ll_N T^{-N} \quad \text{for every } N, \quad (8.19)$$

then, for every $D > 0$,

$$\sup_{K_0} |f_T| \ll_{D, K_0} T^{-D}. \quad (8.20)$$

Proof. Join a ball compactly contained in U to K_0 by finitely many overlapping balls in Ω . Repeated use of the three-ball, or two-constants, inequality gives

$$|f_T| \leq (T^{-N})^\omega (T^{C_2})^{1-\omega}$$

on K_0 , where $\omega > 0$ depends only on the fixed geometry. Take $N > (D + (1 - \omega)C_2)/\omega$. $\square$

8.2. Joint removability at shift collisions. We place the collision algebra before the quantitative continuation that uses it. The next lemma is formal local algebra and contains no analytic continuation input.

Lemma 8.2 (Three-divisor cancellation). *Let x, y be independent local holomorphic coordinates and put $z = x + y$. Suppose a, b, c, H_0 are holomorphic and*

$$F = \frac{a}{xz} + \frac{b}{yz} + \frac{c}{xy} + H_0, \quad a + c \in (x), \quad b + c \in (y), \quad a - b \in (z). \quad (8.21)$$

Then F is holomorphic across $xyz = 0$.

Proof. Over the common denominator, the numerator is

$$N = ay + bx + cz. \quad (8.22)$$

Modulo x , use $z = y$ to obtain $N = (a + c)y$; modulo y , use $z = x$ to obtain $N = (b + c)x$; modulo z , use $y = -x$ to obtain $N = (b - a)x$. The elements x, y, z are pairwise nonassociate prime elements in the local analytic UFD (the corresponding ideals need not be comaximal). Since each of these prime elements divides N , unique factorization gives $xyz \mid N$. $\square$

For the fixed- s assembly, take the coordinates in (8.12). After extracting the simple polar zeta factors, the direct diagonal, reflected dual diagonal, and zero mode have denominator patterns xz , yz , and xy . The three residue cancellations in (8.7)–(8.9) give exactly the three divisibilities in (8.21). Explicitly, $\zeta(1 + \xi) = \xi^{-1} + h_+(\xi)$ and $\zeta(1 - \xi) = -\xi^{-1} + h_-(\xi)$ with $h_\pm$ holomorphic. Equality of the complete Euler residues on $x = 0$, $y = 0$, and $z = 0$ therefore says, respectively, that the corresponding numerator sums

lie in (x) , (y) , and (z) ; these are identities in the local analytic ring, not merely cancellations at generic points of the three hyperplanes. Thus the complete fixed- s integrand is jointly holomorphic even at double or triple intersections. Its separate $1/s$ factor remains, so the prescribed $s = 0$ residue is retained.

For the permutation sum, the collision pairs are

hyperplane	first Euler product	matching Euler product
$\lambda = -\gamma$	$Z_{-\gamma,\beta,\gamma}$	$Z_{-\gamma,\beta,\gamma}, X_{-\gamma,\gamma} = 1$
$\lambda = \beta$	$Z_{-\gamma,\beta,-\beta}$	$Z_{\beta,-\gamma,-\beta}$, equal by first-shift symmetry
$\beta = -\gamma$	$Z_{\lambda,-\gamma,\gamma}$	$Z_{\lambda,-\gamma,\gamma}, X_{-\gamma,\gamma} = 1$

In each row the polar zeta arguments are opposite local coordinates, so the residues have opposite signs, including the finite factors at primes dividing hk .

Proposition 8.3. *The complete expression $\mathcal{P}_\lambda(h, k; t)$ in (8.1) is jointly holomorphic across all shift-collision divisors.*

Proof. Apply Lemma 8.2 with

$$x = \lambda + \gamma, \quad y = -\beta - \gamma, \quad z = \lambda - \beta = x + y. \quad (8.23)$$

The middle, last, and first terms of (8.1) supply the denominator patterns xz , yz , and xy , respectively. The three rows of the preceding table are precisely the required divisibilities. This also handles the common collision locus $\lambda = \beta = -\gamma$, rather than only generic points of the individual hyperplanes. $\square$

This proposition uses only (8.1), the Euler-factor symmetry in (2.5), and $X_{-\gamma,\gamma} = 1$. In particular, it is now proved before, and does not import any estimate from, the quantitative continuation below.

We next justify uniformly the power-factor form quoted after the main theorem. Put

$$p_{A,C_1}(t) := \left(\frac{t}{2\pi}\right)^{-A-C_1} \quad (8.24)$$

and let $\mathcal{P}_\lambda^{\text{pow}}$ be (8.1) with $X_{\lambda,\gamma}$ and $X_{\beta,\gamma}$ replaced by $p_{\lambda,\gamma}$ and $p_{\beta,\gamma}$.

Proposition 8.4 (Collision-safe Stirling replacement). *Uniformly in the shift range (2.10), including all pairwise and triple collisions,*

$$\mathcal{P}_\alpha(h, k; t) - \mathcal{P}_\alpha^{\text{pow}}(h, k; t) \ll_{C,\varepsilon} T^{-1}(\log T)^{C_C} (hk)^\varepsilon \quad (t \in \text{supp } w). \quad (8.25)$$

Consequently its coefficient-weighted h, k, t contribution is $O_{C,\varepsilon}(T^\varepsilon(HK)^{1/2})$.

Proof. Use the collision coordinates

$$x = \lambda + \gamma, \quad y = -\beta - \gamma, \quad z = \lambda - \beta = x + y,$$

and write

$$\delta_1 = X_{\lambda,\gamma} - p_{\lambda,\gamma}, \quad \delta_2 = X_{\beta,\gamma} - p_{\beta,\gamma}.$$

The complete replacement difference is

$$\mathcal{D}_X = \delta_1 Z_{-\gamma,\beta,-\lambda}(h, k) + \delta_2 Z_{\lambda,-\gamma,-\beta}(h, k). \quad (8.26)$$

Set

$$A_2 = xzZ_{-\gamma,\beta,-\lambda}(h,k), \quad A_3 = yzZ_{\lambda,-\gamma,-\beta}(h,k).$$

For $p^\nu \parallel h_0$, put $X_p = p^{-q}$ and $Y_p = p^{-v}$. The finite Euler factor in (2.5) is, initially for $|X_p|, |Y_p| < 1$,

$$\begin{aligned} B_\nu(q,v) &:= (1 - X_p)(1 - Y_p) \sum_{i+j \geq \nu} X_p^i Y_p^j \\ &= 1 - (1 - X_p)(1 - Y_p) \sum_{i+j < \nu} X_p^i Y_p^j \\ &= X_p^\nu + (1 - X_p) \sum_{j=0}^{\nu-1} X_p^j Y_p^{\nu-j}. \end{aligned} \tag{8.27}$$

The middle expression continues B_ν as a symmetric polynomial in X_p, Y_p . Consequently (2.5) shows directly that A_2 and A_3 are holomorphic near the collision locus. Since both the exact and power factors equal 1 when their two shifts sum to zero,

$$\delta_1 \in (x), \quad \delta_2 \in (y).$$

On $z = 0$ one has $\lambda = \beta$, hence $\delta_1 = \delta_2$; moreover the finite Euler factors in A_2 and A_3 agree there by $B_\nu(q,v) = B_\nu(v,q)$ in (8.27), equivalently by the first-shift symmetry in (2.5). Since $y = -x$ on this divisor, the polar factors agree there as the exact identities

$$z\zeta(1-z)|_{z=0} = -1, \quad z\zeta(1+z)|_{z=0} = 1, \quad y\zeta(1+y) = -x\zeta(1-x).$$

Consequently $A_2 = A_3$ on $z = 0$. Thus

$$\delta_1 A_2 \in (x), \quad \delta_2 A_3 \in (y), \quad \delta_1 A_2 - \delta_2 A_3 \in (z).$$

Applying Lemma 8.2 to

$$\mathcal{D}_X = \frac{\delta_1 A_2}{xz} + \frac{\delta_2 A_3}{yz}$$

proves joint holomorphy, including the triple collision.

It remains to retain the factor T^{-1} uniformly. Let $L = \log T$ and choose fixed radii $r_y > r_x > 2C$ and $r_\gamma > C$. The target shift region is contained in the polydisc

$$|x| \leq r_x/L, \quad |y| \leq r_y/L, \quad |\gamma| \leq r_\gamma/L.$$

On its distinguished boundary, $|z| = |x + y| \geq (r_y - r_x)/L$, so no polar divisor is met. Uniform Stirling gives $\delta_j \ll_C T^{-1}$ there. The local formula (2.5) gives

$$Z_{-\gamma,\beta,-\lambda}(h,k), Z_{\lambda,-\gamma,-\beta}(h,k) \ll_{C,\varepsilon} L^{C_\varepsilon} (hk)^\varepsilon;$$

indeed, the final expression in (8.27) shows that, after the two displayed zeta factors are removed, a prime power $p^\nu \parallel h_0$ contributes

$$\ll_C (\nu + 1)p^{-\nu/2 + O_C(\nu/L)},$$

and the exterior k_0 -power is no larger. Hence $\mathcal{D}_X \ll T^{-1} L^{C_C} (hk)^\varepsilon$ on the distinguished boundary. The maximum principle in the three variables (x, y, γ) transfers this bound to the whole polydisc and proves (8.25). Finally

$$\int_{\mathbb{R}} |w(t)| dt \ll T, \quad \sum_{h \leq H} \sum_{k \leq K} \frac{|a_h b_k|}{\sqrt{hk}} (hk)^\varepsilon \ll_\varepsilon T^\varepsilon (HK)^{1/2},$$

which proves the asserted total error. $\square$

8.3. Quantitative continuation of the assembled error.

Proposition 8.5 (Full-kernel assembly continuation). *Under (2.8), for every fixed $C_0, D > 0$,*

$$\mathcal{A}_T(A, B, C_1) - \mathcal{R}_T(A, B, C_1) - \mathcal{D}_T(A, B, C_1) = O_D(T^{-D}) \quad (8.28)$$

uniformly for

$$|A| + |B| + |C_1| \leq C_0 / \log T, \quad (8.29)$$

including every shift collision.

Proof. Put $L = \log T$ and use the scaled variables

$$(\mathbf{a}, \mathbf{b}, \mathbf{c}) = L(A, B, C_1). \quad (8.30)$$

We first fix all geometry. The admissible contour choices

$$c_u = \frac{1}{8}, \quad \varepsilon_s = \frac{1}{8}, \quad \sigma_s = -\frac{3}{8}, \quad c_s = \frac{1}{4} \quad (8.31)$$

satisfy (3.10), (8.10), and, for sufficiently large T , (8.11). For the prescribed C_0 , set

$$\begin{aligned} R_* &= C_0 + 6, \\ \Omega &= \{(\mathbf{a}, \mathbf{b}, \mathbf{c}) : |\mathbf{a}|, |\mathbf{b}|, |\mathbf{c}| < R_*\}, \\ \Omega^\sharp &= \{(\mathbf{a}, \mathbf{b}, \mathbf{c}) : |\mathbf{a}|, |\mathbf{b}|, |\mathbf{c}| < R_* + 1\}. \end{aligned} \quad (8.32)$$

The target compact set

$$K_0 = \{(\mathbf{a}, \mathbf{b}, \mathbf{c}) : |\mathbf{a}| + |\mathbf{b}| + |\mathbf{c}| \leq C_0\} \quad (8.33)$$

has a fixed collar in Ω . We use the fixed open good-patch subset

$$U = \left\{ |\mathbf{a}| < \frac{1}{8}, |\mathbf{b} - 2| < \frac{1}{8}, |\mathbf{c} + 2| < \frac{1}{8} \right\}. \quad (8.34)$$

Its closure lies in Ω . The inverse-scaled projection $(\mathbf{b}/L, \mathbf{c}/L)$ of U lies in (6.15) with $\kappa_0 = 1$ and $c_1 = 5$, while the full set U also satisfies the initial three-variable size condition stated after (6.15), since on U one has $\Re \mathbf{b} > 15/8$, $\Re \mathbf{c} < -15/8$, and $|\mathbf{a}| + |\mathbf{b}| + |\mathbf{c}| < 35/8$.

Take T large enough that

$$L > 64(R_* + 1). \quad (8.35)$$

On $\Omega^\sharp$ the two linear denominator factors in (3.2) then have modulus at least $7/8$, and hence their squares have modulus at least $49/64$. With $C_* =$

$2(R_* + 1)$, (8.31) satisfies (8.11). These are fixed contours throughout $\Omega^\sharp$; the unused kernel divisors stay a fixed positive distance away.

We now prove a polynomial majorant separately for each assembled object. In the convention of (2.2)–(2.3), all constants below may depend on R_*, B_0, M, ε , on the coefficient-bound constants, and on the indicated finite seminorms, but not on H, K, T or the point of the scaled shift domain. No new weight-function dependence is introduced in this section. Choose the auxiliary divisor-bound exponent sufficiently small. Once M is fixed, all factors $L^{C_M} T^{O(\varepsilon)}$ are absorbed into one additional power of T . This last coarse absorption is used only for the polynomial majorant needed by quantitative continuation, not for the sharp T^ε in (2.11).

For the exact-cutoff zero-mode part of $\mathcal{A}_T$, the mesh bound (6.17) and the full-kernel product bound give (6.18). Uniformly on $\Omega^\sharp$,

$$\int_{1/2}^\infty l^{-1/2-\Re A} (1 + l/Y)^{-P} dl \ll_{R_*, B_0} Y^{1/2+(R_*+1)/L} \ll T^{B_0/2}, \quad (8.36)$$

$$\begin{aligned} & \sum_{m \geq 1} m^{-1/2-\Re B} \left\{ \int_{1/2}^\infty n^{-1/2-\Re C_1} (1 + mn/T)^{-Q} dn \right. \\ & \quad \left. + (1 + m/T)^{-Q} \right\} \ll_{R_*} T^{1/2} L. \end{aligned} \quad (8.37)$$

Here P, Q are fixed sufficiently large product-decay orders. To see the second estimate, split at $m = T$. For $m \leq T$ the integral term is at most

$$T^{1/2-\Re C_1} \sum_{m \leq T} m^{-1-\Re B+\Re C_1} \ll_{R_*} T^{1/2} L,$$

because $\sum_{m \leq T} m^{-1+2(R_*+1)/L} \ll e^{2(R_*+1)L}$; increasing P, Q gives arbitrary convergence for $m > T$. This is an exact-cutoff source bound, not a claim of endpoint-deleted normal convergence in the opposite half-space.

The outside coefficients cost at most

$$\sum_{h \leq H} \sum_{k \leq K} \frac{|a_h b_k|}{\sqrt{hk}} \ll_\varepsilon T^\varepsilon (HK)^{1/2} \ll T^{1/2+\varepsilon}, \quad (8.38)$$

where $HK \leq T^{1-\eta}$ follows from (2.8) and $H \geq 1$. Together with the $O(T)$ length of the t -integral, (8.36)–(8.38) give $T^{B_0/2+2+o(1)}$ for each exact zero mode. On the positive line $\Re s = c_s = 1/4$, the two source diagonals are absolutely convergent and the Euler formula gives the deliberately cruder estimate

$$L^{C_M} T^{1+c_s+B_0 c_u} H^{1+c_s+\varepsilon} K^{1/2+\varepsilon} \ll T^{B_0/8+4}. \quad (8.39)$$

The exact factors $X_{B, C_1}(t)$ are $O_{R_*}(1)$ on the scaled domain. The locally uniform exact-cutoff majorants prove holomorphy and

$$\sup_{\Omega^\sharp} |\mathcal{A}_T| \ll T^{B_0/2+4}. \quad (8.40)$$

For $\mathcal{R}_T$, retain the complete sum $\mathcal{P}_{A+u}$ before taking absolute values. On $\Re u = c_u = 1/8$, the collision coordinates

$$x = A + u + C_1, \quad y = -B - C_1, \quad z = A + u - B = x + y$$

satisfy $|x|, |z| \geq 3/32$ on $\Omega^\sharp$ by (8.35). Thus only $y = 0$ can meet this integration line. Away from a fixed scaled tube around $y = 0$, the Euler formula (2.5), Stirling, and vertical-strip zeta bounds give

$$\mathcal{P}_{A+u}(h, k; t) \ll_{R_*, M, \varepsilon} L^{C_M} (hk)^\varepsilon h^{1/2+\varepsilon} (1 + |\Im u|)^{C_M}. \quad (8.41)$$

In the scaled tube $|Ly| < 1/2$ we first use the logically prior algebraic input Proposition 8.3, with the acyclic dependency recorded above. More explicitly, fix A, C_1, u, t and put

$$f(w) = \mathcal{P}_{A+u}(h, k; t)|_{B=-C_1-w/L}.$$

The complete function f is holomorphic for $|w| < 1$ and the circle $|w| = 1/2$ lies inside the fixed collar of $\Omega^\sharp$. The maximum principle bounds every $f(w)$ with $|w| \leq 1/2$ by the maximum on this circle. There $|B + C_1| = 1/(2L)$, so the individual Euler estimates cost only a fixed power of L , while x and z remain bounded away from zero. A finite cover of $\bar{\Omega}$ proves (8.41) throughout the tube, including $y = 0$, without separately estimating a meromorphic term at the collision. The u, t, h, k integrations now give

$$\sup_{\Omega} |\mathcal{R}_T| \ll L^{C_M} T^{1+B_0/8} H^{1+\varepsilon} K^{1/2+\varepsilon} \ll T^{B_0/8+4}. \quad (8.42)$$

Finally consider $\mathcal{D}_T$. This object is defined only after the direct diagonal, reflected dual diagonal, and zero mode have been assembled and the three small residues have cancelled. On $\Re s = \sigma_s = -3/8$, the two pairs of Euler arguments have real parts

$$\begin{aligned} \Re(1 + A + u + C_1 + s) &= 3/4 + O(R_*/L), \\ \Re(1 + B + C_1 + 2s) &= 1/4 + O(R_*/L), \\ \Re(1 + A + u - B + s) &= 3/4 + O(R_*/L), \\ \Re(1 - B - C_1 + 2s) &= 1/4 + O(R_*/L). \end{aligned} \quad (8.43)$$

They stay a fixed distance from both zero and one. The prime-power formula (8.27) and vertical-strip zeta bounds therefore give, for both Euler products on this line,

$$Z \ll_{R_*, M, \varepsilon} L^{C_M} (hk)^{1/2+\varepsilon} (1 + |\Im u| + |\Im s|)^{C_M}. \quad (8.44)$$

Stirling gives $g(s, t) \ll T^{-3/8} (1 + |\Im s|)^{C_M}$; the factor e^{s^2} is Gaussian and $\Gamma(u)$ has exponential decay. Completing both vertical integrals and the h, k sums yields

$$\sup_{\Omega} |\mathcal{D}_T| \ll L^{C_M} T^{1-3/8+B_0/8} H K T^\varepsilon \ll T^{B_0/8+3}. \quad (8.45)$$

This estimate is for the assembled boundary object; no individual meromorphic beta branch has been continued through a collision.

Define, explicitly,

$$\mathcal{E}_{\text{end},T}(A, B, C_1) := \mathcal{A}_T(A, B, C_1) - \mathcal{B}_T(A, B, C_1) - \mathcal{D}_T(A, B, C_1). \quad (8.46)$$

It follows from (8.40), (8.42), and (8.45) that

$$F_T(\mathbf{a}, \mathbf{b}, \mathbf{c}) := \mathcal{E}_{\text{end},T}(\mathbf{a}/L, \mathbf{b}/L, \mathbf{c}/L), \quad \sup_{\Omega} |F_T| \ll T^{C_2}, \quad C_2 = \frac{B_0}{2} + 5. \quad (8.47)$$

Thus the enlarged scaled domain, its collar, and the polynomial exponent are all fixed independently of T . On U , the good-patch assembly (8.16) gives $\sup_U |F_T| \ll_N T^{-N}$ for every fixed N .

The parameter order is acyclic. First fix $B_0 > 2$, C_0 , and the desired final saving D ; this fixes $K_0 \in \Omega$, $U \in \Omega$, and the propagation exponent $\omega = \omega(C_0) > 0$. Next choose one integer M such that, for $\sigma = M + 1/2$,

$$(B_0 - 1)\sigma > D + B_0 + 20. \quad (8.48)$$

This coarse condition also dominates the later permutation and diagonal-boundary left-line requirements. With this M fixed, $C_2 = B_0/2 + 5$ is fixed; only the lower threshold for T may depend on M, C_0 . Now choose

$$N = \left\lceil \frac{D + (1 - \omega)C_2}{\omega} \right\rceil + 2 \quad (8.49)$$

and afterward the endpoint integration-by-parts and product-tail orders needed for the good-patch $O(T^{-N})$. In particular M is independent of N . Applying Lemma 8.1 to F_T on the fixed geometry $U \in \Omega$, with target K_0 , proves (8.28). $\square$

Only the difference in (8.28) has been propagated. In particular, no power-sized estimate for an individual diagonal boundary term has been continued by unique continuation.

9. FINAL CONTOUR MOVES, COLLISIONS, AND COMPLETION OF THE PROOF

The two u -moves in this section are performed only after Proposition 8.5. They are distinct: first we move the full diagonal boundary, and then the complete three-permutation expression.

9.1. The diagonal-boundary u -move. The ordinary $u = 0$ residue of (8.15) is

$$\begin{aligned} \mathcal{R}_{\text{diag},0}(h, k) = \int_{\mathbb{R}} w(t) \Big\{ & \frac{1}{2\pi i} \int_{(\sigma_s)} \frac{G(s)}{s} g_{\beta,\gamma}(s, t) \\ & \times Z_{\alpha,\beta+s,\gamma+s}(h, k) \, ds \\ & + X_{\beta,\gamma}(t) \frac{1}{2\pi i} \int_{(\sigma_s)} \frac{G(s)}{s} g_{-\gamma,-\beta}(s, t) \\ & \times Z_{\alpha,-\gamma+s,-\beta+s}(h, k) \, ds \Big\} dt. \quad (9.1) \end{aligned}$$

Move each u -line in (8.15) to

$$\Re u = -\sigma_u, \quad \sigma_u = M + \frac{1}{2}. \quad (9.2)$$

On $\Re s = \sigma_s$, the only u -dependent arithmetic poles are

$$u = -\alpha - \gamma - s, \quad u = \beta - \alpha - s. \quad (9.3)$$

Both have real part $1/2 - \varepsilon_s + O((\log T)^{-1}) > c_u$, so they lie to the right of the original line. The pole at $u = 0$ gives (9.1); the potential poles $u = -1, \dots, -M$ vanish by (3.11).

For the new-line estimate, the finite Euler factor in (2.5) may be written

$$B_\nu(q, v) = X^\nu + (1 - X) \sum_{j=0}^{\nu-1} X^j Y_0^{\nu-j}, \quad X = p^{-q}, \quad Y_0 = p^{-v}. \quad (9.4)$$

This follows by summing the geometric tail in

$$(1 - p^{-q})(1 - p^{-v}) \sum_{i+j \geq \nu} p^{-qi-vj}.$$

On (9.2), (9.4) and vertical-strip zeta bounds give

$$Z_{\alpha+u, \beta+s, \gamma+s}(h, k), \quad Z_{\alpha+u, -\gamma+s, -\beta+s}(h, k) \ll_M (\log T)^{C_M} (hk)^\varepsilon h^{\sigma_u+O(\varepsilon_s)+\varepsilon} \\ \times (1 + |u| + |s|)^{C_M}. \quad (9.5)$$

Together with $g(s, t) \ll T^{\sigma_s}(1 + |s|)^{C_M}$, the gamma decay in u , and the Gaussian decay in s , the new lines after the outer coefficient sum are

$$\ll T^{1+\sigma_s+\varepsilon} Y^{-\sigma_u} H^{\sigma_u+O(\varepsilon_s)+\varepsilon} (HK)^{1/2}. \quad (9.6)$$

Since $H \leq T^{2(1-\eta)/3}$ and $Y = T^{B_0}$ with $B_0 > 2$, M can be chosen so that (9.6) is $O_D(T^{-D})$. At $u = 0$ the Euler arguments on $\Re s = \sigma_s$ remain a fixed distance from their poles, and hence

$$\sum_{h \leq H} \sum_{k \leq K} \frac{|a_h b_k|}{\sqrt{hk}} |\mathcal{R}_{\text{diag}, 0}(h, k)| \ll_\varepsilon T^{1/2+\varepsilon} (HK)^{1/2}. \quad (9.7)$$

Consequently

$$\mathcal{D}_T = \sum_{h \leq H} \sum_{k \leq K} \frac{a_h \overline{b_k}}{\sqrt{hk}} \mathcal{R}_{\text{diag}, 0}(h, k) + O_D(T^{-D}). \quad (9.8)$$

9.2. The permutation u -move.

Lemma 9.1 (Left-line bound). *For fixed $\sigma > 0$ and $\Re \lambda = -\sigma + O((\log T)^{-1})$,*

$$Q(u) \mathcal{P}_\lambda(h, k; t) \ll_{\sigma, M} (\log T)^{C_M} (hk)^\varepsilon (T^\sigma + h^\sigma) (1 + |\Im u|)^{C_M}. \quad (9.9)$$

After multiplication by $\Gamma(u)$, this is integrable on the complete vertical line.

Proof. Retain the complete sum $\mathcal{P}_\lambda$ before estimating it. In the collision coordinates

$$x = \lambda + \gamma, \quad y = -\beta - \gamma, \quad z = \lambda - \beta = x + y,$$

the line $\Re \lambda = -\sigma + O((\log T)^{-1})$ keeps x and z a fixed positive distance from zero. Thus only $y = 0$ can meet this line.

Away from $|y| < 1/(2 \log T)$, use (9.4) term by term. If the first shift of Z is λ and its third shift is γ or $-\beta$, the local product is at worst $h_0^{\sigma+\varepsilon}$. If the third shift is $-\lambda$, both Euler arguments have real part $1 + \sigma + O((\log T)^{-1})$, and the local factor cancels the apparent exterior $h_0^{1/2+\sigma}$, leaving $h_0^{-1/2+\varepsilon}$. Stirling gives $X_{\lambda,\gamma}(t) \ll T^{\sigma+\varepsilon}$; the other exact X factor has size $T^{O((\log T)^{-1})}$. Standard vertical-strip zeta bounds then give (9.9), with the possible distance to $y = 0$ absorbed by a fixed power of $\log T$.

In the remaining tube $|y| < 1/(2 \log T)$, apply the joint removability of the *complete* permutation sum from the logically prior Proposition 8.3 before taking absolute values. With γ fixed, write $\beta = -\gamma - w/\log T$. The maximum principle on $|w| \leq 1/2$ bounds every value in the tube by the already established bound on $|w| = 1/2$. That circle changes only the logarithmic factor, and x, z remain uniformly separated from zero. This proves (9.9) throughout the tube, including the collision, without estimating either meromorphic summand there. The vertical-strip zeta bounds and the uniform form of Stirling's formula used above hold for every $\Im u$; the complementary compact ranges of the gamma arguments are bounded directly. Thus (9.9) holds on the complete vertical line, where $\Gamma(u)$ dominates the displayed polynomial in $|\Im u|$. $\square$

Set $\sigma = M + 1/2$ and move the common u -line in (8.17) from c_u to $-\sigma$. The complete residue list is as follows.

- (i) At $u = 0$, $\Gamma(u)$ has residue one and $Q(0) = 1$; this gives $\mathcal{P}_\alpha$.
- (ii) The gamma residues at $u = -j$, $1 \leq j \leq M$, vanish because $R_M(-j) = 0$.
- (iii) The arithmetic collision locations $u = -\alpha - \gamma$ and $u = \beta - \alpha$ are removable in the complete $\mathcal{P}$ by the logically prior Proposition 8.3.
- (iv) The poles $u = 1/2 + 2n - \alpha - it$ of $X_{\alpha+u,\gamma}(t)$ lie to the right of c_u and are not crossed. The $n = 0$ pole encountered in the preliminary move from $\Re u = 1$ to c_u is exponentially small. Denominator gamma factors give zeros.

There are no other poles between the two lines. By Lemma 9.1, the new line for fixed h, k is bounded by

$$T(\log T)^{C_M} (hk)^\varepsilon \{(T/Y)^\sigma + (h/Y)^\sigma\}. \quad (9.10)$$

After the outer sum, choose M so that

$$(B_0 - 1)\sigma > \frac{1}{2}(1 - \eta) + D + 1 + 2\varepsilon. \quad (9.11)$$

Then (9.10) is $O(T^{-D})$ after the outer sum.

9.3. Proof of the main theorem. By (3.13), the coupled smoothed expansion differs from (2.7) by an arbitrary negative power. Its $r = 0$ diagonals and fixed-orientation zero modes satisfy (8.28). The diagonal boundary is reduced by (9.8) and bounded by (9.7). The complete permutation line contributes only its $u = 0$ residue, by (9.10). Finally, the nonzero frequencies are bounded by (5.28). We obtain

$$\begin{aligned} \mathcal{I}_{\alpha,\beta,\gamma}(a,b) &= \sum_{h \leq H} \sum_{k \leq K} \frac{a_h \bar{b}_k}{\sqrt{hk}} \int_{\mathbb{R}} w(t) \mathcal{P}_{\alpha}(h,k;t) dt \\ &\quad + O_{\eta,C,\varepsilon} \left(T^{\varepsilon} \{ KH^{3/2} + T^{1/2} (HK)^{1/2} \} \right). \end{aligned}$$

By Proposition 8.3, the main term is jointly holomorphic at every shift collision. This is (2.11). The replacement (2.12), with its total uniform error, follows from Proposition 8.4; in particular, no individual meromorphic Z -term is estimated at a collision. This completes the proof.

10. THE LIMITING BUI CORNER

This section is a boundary calculation and is not used in the proof of Theorem 2.1. Write Bui's two mollifier lengths as

$$H \leq T^{4/7-\varepsilon_B}, \quad K \leq T^{1/4-\varepsilon_B}. \quad (10.1)$$

Then

$$\frac{3}{2} \left(\frac{4}{7} - \varepsilon_B \right) + \left(\frac{1}{4} - \varepsilon_B \right) = \frac{31}{28} - \frac{5}{2} \varepsilon_B. \quad (10.2)$$

It is less than one precisely when

$$\varepsilon_B > \frac{3}{70}. \quad (10.3)$$

Thus Theorem 2.1 already covers every Bui box with this strict margin.

If the margin is suppressed and only the limiting exponents $(\theta_1, \theta_3) = (4/7, 1/4)$ are compared, the nonzero-mode estimate is

$$KH^{3/2} = T^{31/28+o(1)}. \quad (10.4)$$

To reach an error $T^{1-\delta}$ through the same reduction would require an additional saving of

$$T^{3/28+\delta} \quad (10.5)$$

in the limiting exponent budget. The pointwise completed Weil bound in Lemma 5.1 does not provide it for two arbitrary coefficient sequences.

An averaged reciprocal or Kloosterman-fraction estimate retaining both coefficient sequences could be a sufficient new input for the nonzero frequencies, but no such estimate is assumed or proved here. Nor would it by itself settle every analytic detail at the corner: the endpoint-deletion and full-kernel estimates in Section 6 were proved and audited under (2.8), and their uniformity at the limiting corner has not been established. It is therefore not valid to take a minimum over different Poisson orientations or to describe a single averaged estimate as the unique remaining dependency.

Finally, the cubic product in (2.7) is the “twisted third moment” appearing in Bui’s mollifier cross term. It is not the sixth moment $\int |\zeta(1/2+it)|^6 dt$, and Theorem 2.1 does not imply a sixth-moment asymptotic.

APPENDIX A. PROOF LEDGER AND REPRODUCIBILITY

This appendix records the dependency structure used to check the proof. It is not a substitute for the arguments in the body of the paper.

A.1. Dependency ledger.

ID	claim	location
A1	The unnormalized AFE includes all four remote completed-zeta residues, and the long zeta series is expanded only on $\Re u = 1$.	Lemma 3.1, (3.9)
A2	The coupled weight has simultaneous product decay on complete vertical contours in all four size quadrants, with a central Stirling estimate and a Gaussian complementary region; the same bound is locally uniform on the external A -compact sets used in Section 6.	Lemma 3.2
A3	Extracting $d = (h, km)$ gives a coprime inverse and exact Poisson density d/h .	(4.4)–(4.6)
A4	The fixed- l nonzero frequencies are $O(T^\varepsilon KH^{3/2})$, with the gcd factor in both completion terms averaged by (5.5) and the actual TS Poisson-integral size retained.	Lemma 5.1 and propositions 5.2 and 5.3
A5	The artificial $l < 1$ and $n < 1$ endpoint regions have arbitrary saving on the good half-space; the singular layer has exact gcd cancellation.	Proposition 6.2, (6.23)
A6	All four branches reach genuine Tonelli domains. At finite height the source and destination satisfy the rectangle identity including both horizontal edges; compact-uniform dominators then justify, in order, $V \rightarrow \infty$, $R \rightarrow \infty$, and $\kappa \rightarrow 0^+$. The resulting branchwise continuations are recovered on a common s -contour by a simultaneous A -continuation and moving-line argument that keeps every small arithmetic divisor on the same side. The path is split at remote crossings into pole-free pieces, with the crossings represented by rapidly decreasing successive residues.	(6.59), (6.62), (6.63), (6.65), Lemma 7.4

ID	claim	location
A7	The beta integrals, finite Euler product, and archimedean reflection identities give the evaluated zero modes.	Section 7
A8	Three pairs of fixed- u small s -residues cancel before the diagonal boundary is separated.	(8.7)–(8.9)
A9	Quantitative continuation is applied only to $\mathcal{A}_T - \mathcal{R}_T - \mathcal{D}_T$.	Proposition 8.5
A10	The diagonal-boundary and permutation u -moves retain only their $u = 0$ residues; $-1, \dots, -M$ are killed by R_M .	Section 9
A11	Double and triple shift collisions are jointly removable by local division by three linear factors.	Lemma 8.2 and proposition 8.3
A12	Combining the zero mode, nonzero mode and diagonal boundary gives the error stated in Theorem 2.1.	final paragraph of Section 9
A13	Replacing the exact gamma ratios by their power factors is performed on the complete permutation difference, whose three collision divisors are jointly removable; no meromorphic summand is estimated separately.	Proposition 8.4

The ledger IDs are organizational rather than chronological. The collision statement A11 is proved directly from (8.1), (2.5), (2.6), and local three-divisor division and is placed before the continuation proposition that implements A9. Thus the directed edge is $A11 \rightarrow A9$, with no reverse edge.

A.2. Additional adversarial checks. We record nine stress tests applied to the quantitative and uniformity claims used in the proof.

- D1. The coefficient hypotheses are used through the actual block norms $\mathcal{A}_{H_0}$ and $\mathcal{B}_{K_0}$ in (5.6), or through the explicit outer absolute sum in (8.38). After choosing a smaller auxiliary exponent,

$$\sum_{h \leq H} \sum_{k \leq K} \frac{|a_h b_k|}{\sqrt{hk}} \ll_{\varepsilon} T^{\varepsilon} (HK)^{1/2}.$$

No ℓ^1 coefficient norm is replaced by a supremum without its associated counting factor.

- D2. Every occurrence of T^{ε} is interpreted after choosing the auxiliary divisor and localization exponents sufficiently smaller than the final ε . In particular, (5.19) is obtained only after the fixed completion order is known. The logarithmic dyadic counts and the finitely many subsequent losses may therefore be combined into one final T^{ε} . The

coarser absorption used to prove the polynomial majorant in Proposition 8.5 is not used as part of the sharp error term. The corresponding implicit-constant and weight-seminorm dependencies are exactly those recorded in (2.2)–(2.3); in particular, fixed M permits $(\log T)^{C_M} \ll_{M,\varepsilon} T^\varepsilon$ but not the absorption of any fixed $T^{\delta(M)}$.

- D3. The only size consequences used beyond (2.8) are

$$HK \leq T^{1-\eta}, \quad H \leq T^{2(1-\eta)/3}, \quad K \leq T^{1-\eta},$$

which follow from $H, K \geq 1$. These are sufficient for (6.7), (6.9), the polynomial majorants, and the two final left-line estimates. No ordering between H and K , or stronger hidden length condition, is used.

- D4. All differentiation and integration-by-parts orders are fixed before T . Finite completion uses only a fixed number of derivatives, while the endpoint orders depend on the prescribed fixed saving. The parameter order

$$(B_0, C_0, D) \longrightarrow M \longrightarrow N \longrightarrow (J, P, Q)$$

is recorded in (8.48)–(8.49). Consequently only finitely many seminorms from (2.1) enter the implied constants.

- D5. After scaling the shifts by $\log T$, the domains (8.32) and their enlargement have a fixed collar, and the contours in (8.31) are independent of T and of the point in the target compact set. The threshold (8.35) keeps the kernel divisors and the relevant moving poles a fixed distance from these contours. Thus no continuation argument relies on a collar shrinking with the shifts.

- D6. Uniformity is checked simultaneously at

$$\alpha + \gamma = 0, \quad \beta + \gamma = 0, \quad \alpha - \beta = 0,$$

including the common locus $\alpha = \beta = -\gamma$. The local coordinates in (8.23) satisfy $z = x + y$, and Lemma 8.2 proves divisibility by the full product xyz , rather than cancellation only at generic points. Fixed-radius Cauchy estimates on the scaled domain preserve the stated uniform bounds at the filled-in values.

- D7. Individual meromorphic terms are not estimated on a collision divisor and then combined formally. The complete permutation sum is retained before (8.41) is applied; the direct diagonal, reflected dual diagonal, and zero mode are assembled before the boundary estimate (8.44); and the source object is bounded from its exact-cutoff representation. Cancellation therefore precedes every absolute-value estimate that crosses a collision.
- D8. At the source side of Proposition 6.4 the regulator is removed in the physical measure $\sum_{m,\rho} dv$. The bulk part uses (6.17). Before any integration or summation, however, the compact parameter supremum is bounded by the eight endpoint corners in (6.35). The genuine $n = 0$ and $l = 0$ crossings are then integrated in their physical variables by

(6.37) and (6.39); the two noncrossing endpoint pieces are summed separately. No supremum in an independent endpoint variable is used. The false general sampling assertion has therefore been removed.

- D9. The final error ledger contains both surviving contributions: (5.28) gives $T^\varepsilon K H^{3/2}$, and (9.7) gives $T^{1/2+\varepsilon}(HK)^{1/2}$. The AFE recovery error, the continued endpoint error, and both regulator left lines are smaller than any prescribed fixed power by (8.28), (9.8), and (9.10). Thus the terms retained in (2.11) cover the complete error decomposition.

Scope of the audit. No unsupported inference remains in these nine internal checks after the source supremum was replaced by its pointwise corner majorant and the power replacement was proved on the complete holomorphic difference. Fresh-context specialist agents separately rechecked the source bridge, the contour/Gamma argument, and the global dependency graph. The resulting classification is candidate complete, not formally verified and not an external human peer-review certificate. The finite numerical tests likewise check only explicit identities.

A.3. Finite identity checks. The accompanying script `verification/check_zero_identities.py` uses arbitrary-precision arithmetic to test the two gamma identities and the linear approach to all three collision divisors and to their common triple intersection. A reference run at 80 decimal digits produced relative errors between approximately 10^{-52} and 10^{-80} for the gamma identities. In the collision tests, decreasing the displacement from 10^{-12} to 10^{-16} to 10^{-20} decreased the successive differences by a factor of 10^{-4} at each step. These are finite consistency checks only; the contour deformations, Tonelli arguments, and continuation are proved analytically in the body of the paper. The same script now applies these collision tests to the complete exact-minus- power replacement difference and checks its first Stirling saving under $T \mapsto 2T$. The companion script `verification/check_regulator_bridge.py` checks the four beta identities and a truncated physical endpoint integral converging to its finite limit; it does not attempt to certify the compact-uniform analytic dominator.

A.4. Release integrity. The source folder distributed with this paper contains the complete manuscript source, bibliography, compiled PDF, numerical verification scripts, and their fixed-version arbitrary-precision dependency and license. A legacy `MANIFEST.sha256`, when retained with an imported baseline, records only that baseline; it is neither required nor presented as a digest of the revised folder. Each verification script uses explicit release thresholds and prints `status=PASS` only after all of its checks have completed.

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